

# Data Analytics (and Data Driven Decision)

## Descriptive and Predictive Analytics



UNIVERSITÀ  
DEGLI STUDI  
DELL'AQUILA



DISIM  
Dipartimento di Ingegneria  
e Scienze dell'Informazione  
e Matematica

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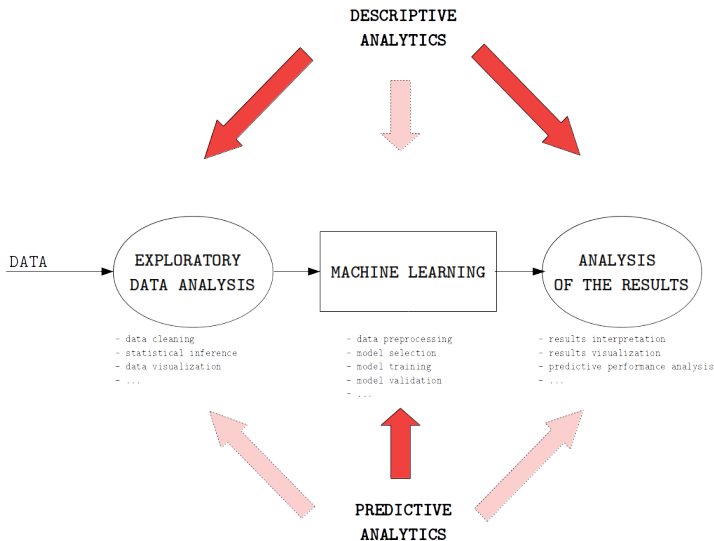


# Overview

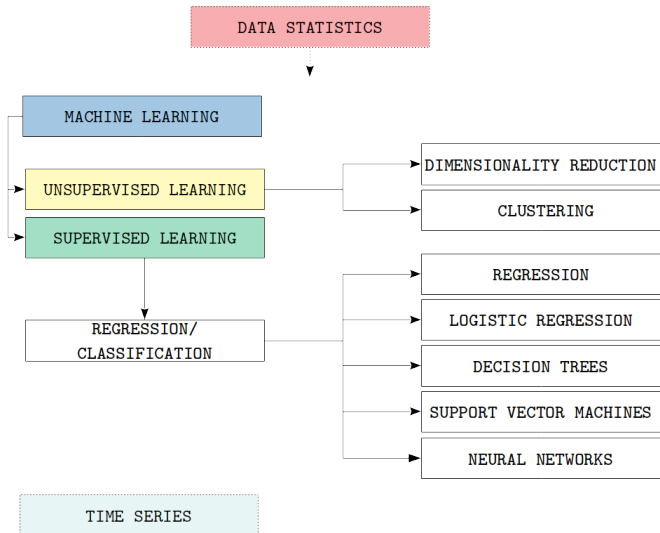
- 1 Introduction
- 2 Basic statistics
- 3 Multivariate statistics and Principal Component Analysis
- 4 Unsupervised learning: Clustering
- 5 Supervised learning
- 6 Overview on Time Series

# Data Analytics: Descriptive and Predictive Analytics

a typical data driven decision process ...



# Course in a glance



Machine Learning is about learning from **data** (predictive models)

- data collected, manipulated and cleaned
- good machine learning algorithms are used
- not enough...
  - ▶ data are **partial** (sampled) and **noisy** (errors)
  - ▶ big data: properly sampled? unbiased?

statistics needed with data!

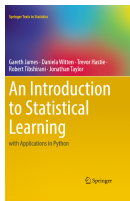
introductory part of the course on **basic statistics**

# General course information

- material used during lectures (slides, exercises) is self-contained (sufficient for the exam preparation)
- reference books

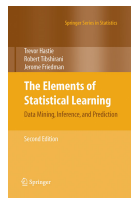
'An Introduction to Statistical Learning'

[www.statlearning.com](http://www.statlearning.com)



'The Elements of Statistical Learning'

[www.web.stanford.edu/~hastie/ElemStatLearn](http://www.web.stanford.edu/~hastie/ElemStatLearn)



- (annotated) slides and exercises published on the Teams course channel
- exercises are Python notebooks (can be replicated with any software/tool you prefer (Matlab, Excel, R, ...))
- **office hours** by appointment to be arranged by mail:  
monday 11:00-13:00, friday 11:00-13:00



# Exam information

Data Analytics determines 50% of the final mark

- **written exam** (up to 30 points):
  - ▶ multiple choice questions test
  - ▶ open-ended questions test
- **mandatory group project** (up to 1 point):  
implementation and presentation of a Data Analytics work on a dataset provided by the lecturers

# Overview

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# Basic statistics concepts

arguments of this part

- DISTRIBUTIONS: SPREAD AND SIZE
- RANDOM VARIABLES
- SAMPLING
- INFERENCE
- RESAMPLING

# Distributions: spread and size

how a **variable** is distributed in a **population**

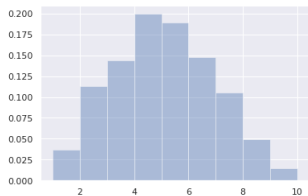
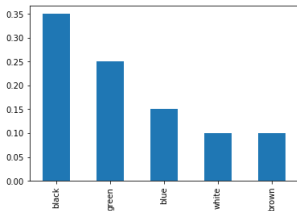


statistic is about synthesis, how to synthesize distributions?

different representations according to

- type of variables: continuous, categorical, ordered,...
- variable values, frequencies,...
- ...

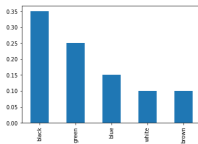
| idx | value |
|-----|-------|
| 1   | 6.86  |
| 2   | 0.62  |
| 3   | 4.53  |
| 4   | 9.95  |
| 5   | 8.88  |
| 6   | 9.58  |
| 7   | 3.62  |
| 8   | 7.06  |
| 9   | 5.29  |
| 10  | 4.65  |
| 11  | 9.16  |
| 12  | 1.32  |
| 13  | 4.97  |
| 14  | 5.72  |
| 15  | 4.47  |
| 16  | 8.56  |
| 17  | 3.58  |
| 18  | 9.69  |
| 19  | 6.87  |
| 20  | 9.31  |



a distribution is a way of representing how a variable is distributed in a population

which measures are important in a distribution?

| idx | value |
|-----|-------|
| 1   | 6.86  |
| 2   | 0.62  |
| 3   | 4.53  |
| 4   | 9.95  |
| 5   | 8.88  |
| 6   | 9.58  |
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| 15  | 4.47  |
| 16  | 8.56  |
| 17  | 3.58  |
| 18  | 9.69  |
| 19  | 6.87  |
| 20  | 9.31  |



**spread** (variability): how much the values of the variable vary in the population?

given a population of  $n$  individuals (units, occurrences, events,...) with (observed) values  $x_1, x_2, x_3, \dots, x_n$ , how to measure variability?

- total variability  $\sum_{i=1, n} \sum_{j=2, n, j > i} |x_i - x_j|$
- average variability  $\frac{1}{n(n-1)} \sum_{i=1} \sum_{j=2, n, j > i} |x_i - x_j|$

spread based on **differences among values**

another way to measure spread: **distances with a reference value 'a'**

$$\sqrt[d]{\frac{1}{n} \sum_{i=1,n} |x_i - a|^d} \quad \text{or} \quad \frac{1}{n} \sum_{i=1,n} |x_i - a|^d \quad (1)$$

- (1) grows with the differences between values  $x_i$  and  $a$
- (1) represents an overall distance measure between the values and the reference

once chosen a distance how to determine the reference value (**center**) w.r.t. that distance?

- center is something intermediate between values...
- center is not too far from any of the values...

$$c_d = \arg \min_{a \in \mathcal{R}} \sqrt[d]{\frac{1}{n} \sum_{i=1,n} |x_i - a|^d} \quad \text{or} \quad \arg \min_{a \in \mathcal{R}} \frac{1}{n} \sum_{i=1,n} |x_i - a|^d \quad (2)$$

if we replace each value  $x_i$  with  $c_d$  the loss of information is minimized!

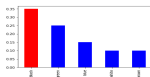


each distance has its center: the **size** (centrum,dimension) of the distribution

•  $\arg \min_{a \in \mathbb{R}} \frac{1}{n} \sum_{i=1,n} |x_i - a|^2$  **MEAN**  $\mu = \frac{1}{n} \sum_{i=1,n} x_i$

•  $\arg \min_{a \in \mathbb{R}} \frac{1}{n} \sum_{i=1,n} |x_i - a|^1$  **MEDIAN**  $\xleftarrow{50\%} \bullet \xrightarrow{50\%}$

•  $\arg \min_{a \in \mathbb{R}} \frac{1}{n} \sum_{i=1,n} |x_i - a|^0$  **MODE**



spread + size = good synthesis of the distribution!

summing up...

POPULATION OF  $n$  INDIVIDUALS  
WITH VALUES  $x_i, i=1, \dots, n$  (continuous, categorical, ...)

CHOOSE THE DISTANCE  
(SPREAD MEASURE)

$$\begin{aligned} &|x_i - a|^2 \\ &|x_i - a|^1 \\ &|x_i - a|^0 \end{aligned}$$

COMPUTE THE SIZE

MEAN  
MEDIAN  
MODE

## EXAMPLE: 2 different populations...

| idx | age |
|-----|-----|
| 1   | 35  |
| 2   | 36  |
| 3   | 34  |
| 4   | 35  |
| 5   | 33  |
| 6   | 37  |
| 7   | 36  |
| 8   | 31  |
| 9   | 38  |
| 10  | 35  |

Figure: population 1: mean=35

| idx | age |
|-----|-----|
| 1   | 68  |
| 2   | 70  |
| 3   | 65  |
| 4   | 71  |
| 5   | 67  |
| 6   | 1   |
| 7   | 2   |
| 8   | 2   |
| 9   | 3   |
| 10  | 1   |

Figure: population 2: mean=35

the spread values are 2 and 35.0365!



most used measures for **continuous variables**

- SIZE

$$\mu = \frac{1}{n} \sum_{i=1,n} x_i$$

- SPREAD

- ▶ VARIANCE

$$\sigma^2 = \frac{1}{n} \sum_{i=1,n} (x_i - \mu)^2$$

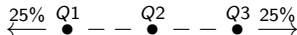
- ▶ STANDARD DEVIATION

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1,n} (x_i - \mu)^2}$$

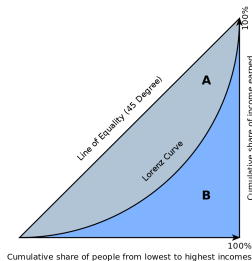
for categorical, ordinal... ?

## other SPREAD measures

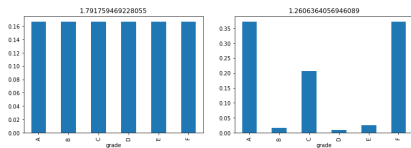
- INTERQUARTILE RANGE (IQR):  $Q3 - Q1$



- GINI COEFFICIENT:  $G = \frac{\sum_{i=1,n} \sum_{j=1,n} |x_i - x_j|}{2n \sum_{i=1,n} x_i}$



- ENTROPY:  $H = - \sum_{i=1,n} f_i \log(f_i)$



## exercise 1.1

VARIANCE ( $\sigma^2 = \frac{1}{n} \sum_{i=1,n} (x_i - \mu)^2$ ) has an alternative formula

- $\sigma^2 = \frac{1}{n} \sum_{i=1,n} x_i^2 - \left(\frac{1}{n} \sum_{i=1,n} x_i\right)^2 = E[x^2] - E[x]^2$

where  $E[\cdot]$  is the expected value (average)

- $\sigma^2 = \frac{1}{n} \sum_{i=1,n} (x_i - \mu)^2 = \frac{1}{n} \sum_{i=1,n} (x_i^2 + \mu^2 - 2x_i\mu) =$

$$\frac{1}{n} \sum_{i=1,n} x_i^2 + \frac{1}{n} \sum_{i=1,n} \mu^2 - \frac{2\mu}{n} \sum_{i=1,n} x_i = \frac{1}{n} \sum_{i=1,n} x_i^2 + \frac{1}{n} \sum_{i=1,n} \mu^2 - 2\mu^2 =$$

$$\frac{1}{n} \sum_{i=1,n} x_i^2 - \mu^2 - \mu^2 + \frac{1}{n} \sum_{i=1,n} \mu^2 = E[x^2] - \mu^2 - \mu^2 + \frac{1}{n} n \mu^2 =$$

$$E[x^2] - \mu^2 = E[x^2] - E[x]^2 = \sigma^2$$

alternative formula convenient for incremental computations!

## exercise 1.2



motivating game:

- I observe a variable  $n$  times and hide the results to you, then compute mean and variance and let you know only these 2 values
- then a new value of the same variable is observed.

Based on the information you have

- 1 what do you expect this new value to be ?
- 2 should you bet on it, how much money would you bet ?

# Random Variables

what is a **random variable**?

It is a variable that can assume different values, for which it is known a **probability distribution** over such values (**states**)

probability distribution is how the random variable is distributed over its states when we observe it many time

probability distribution of a random variable measures the "likelihood" of its states

example of a random variable:

$$\text{TOSS A COIN...} P(H) = P(T) = 0.5 = p$$

# BINOMIAL DISTRIBUTION

- Bernoulli process: a sequence of observations of a binary random variable where each observation has probability of success  $p$  and of failure  $1 - p$  with  $p \in (0, 1)$
- Binomial distribution:  
**discrete** probability distribution of the number of successes  $k$  of  $n$  independent observations of a Bernoulli process

$$\binom{n}{k} p^k (1 - p)^{n-k}$$

the probability of getting  $k$  heads (or tails) by tossing a coin  $n$  times

- the probability of  $n$  independent events is the **product** of the probabilities of the single events

$$\prod_{i=1}^k p \prod_{i=1}^{n-k} (1-p) = p^k (1-p)^{n-k}$$

- how many possible way of getting  $k$  successes on  $n$  observations (trials)?

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

example:  $k = 2, n = 4 \rightarrow \binom{4}{2} = 6$ :

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | H | H | T | T |
| 2 | T | H | H | T |
| 3 | T | T | H | H |
| 4 | H | T | T | H |
| 5 | H | T | H | T |
| 6 | T | H | T | H |

exercise 1.3

given a set  $\Omega$  of all possible values  $x_i$  of a discrete random variable  $X$ , and a probability  $p_i$  associated to each  $x_i$  according to a probability distribution

- MEAN, EXPECTED VALUE, PREDICTION

$$m = \mathbb{P}(X) = \sum_{i \in \Omega} p_i x_i$$

- VARIANCE

$$\text{var}(X) = \sum_{i \in \Omega} p_i (x_i - m)^2$$



## continuous probability distributions

- **density** function  $f(x)$ :

$$f(x) \geq 0, f(x) \leq 1,$$

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

- **cumulative** function  $F(x)$  :

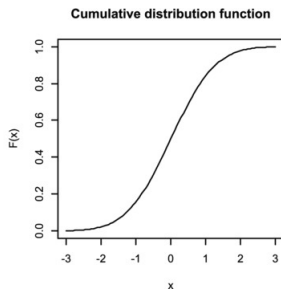
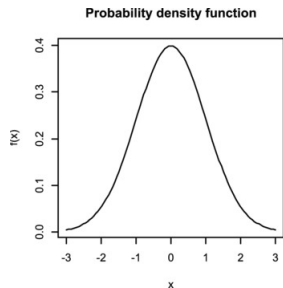
$$F(x) = \int_{-\infty}^x f(y)dy$$

what is the probability that...

...  $x = a$ ?

...  $x \leq b$ ?

...  $a \leq x \leq b$ ?



given a continuous random variable  $X$  with values  $x$  defined on  $[-\infty, \infty]$  associated to a continuous probability density function  $f(x)$

- MEAN, EXPECTED VALUE, PREDICTION

$$m = \mathbb{P}(X) = \int_{-\infty}^{\infty} xf(x)dx$$

- VARIANCE

$$\text{var}(X) = \int_{-\infty}^{\infty} (x - m)^2 f(x) dx$$

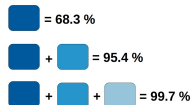
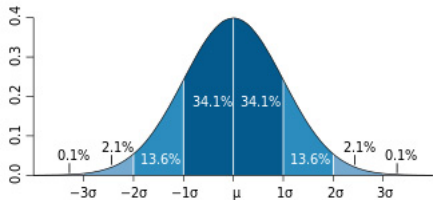
## UNIFORM DISTRIBUTION $U_{a,b}(x)$

- defined on a interval  $[a, b]$  with  $a < b$
- the probability of any value in the interval is the same (outside is 0)

$$f(x) = U_{a,b}(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

- probability of an interval of lenght  $h$ :  $\frac{h}{b-a}$

## NORMAL DISTRIBUTION $N_{\mu,\sigma}(x)$



- defined on the interval  $[-\infty, \infty]$

density:

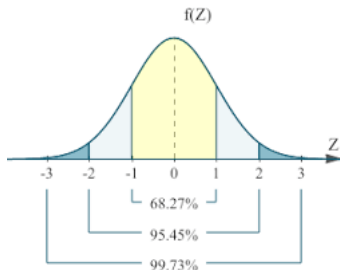
$$f(x) = N_{\mu,\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

cumulative:

$$F(x) = \Phi_{\mu,\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2} dy$$

- symmetric
- fully described by parameters  $\mu$  (mean) and  $\sigma$  (standard deviation)
- central value = mean, median, mode

# STANDARD NORMAL DISTRIBUTION $N(x)$ (often denoted as $N(Z)$ )



- $\mu = 0$  and  $\sigma = 1$

density:

$$N(Z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}Z^2}$$

cumulative:

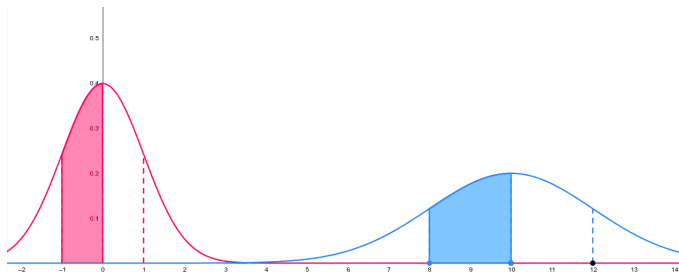
$$\Phi(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^Z e^{-\frac{1}{2}y^2} dy$$

- for  $\Phi(Z)$  available a **table of values**

$$\Phi(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^Z e^{-\frac{1}{2}y^2} dy$$

| <i>z</i> | 0      | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |
|----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| +0       | .50000 | .50399 | .50798 | .51197 | .51595 | .51994 | .52392 | .52790 | .53188 | .53586 |
| +0.1     | .53983 | .54380 | .54776 | .55172 | .55567 | .55966 | .56360 | .56749 | .57142 | .57535 |
| +0.2     | .57926 | .58317 | .58706 | .59095 | .59483 | .59871 | .60257 | .60642 | .61026 | .61409 |
| +0.3     | .61791 | .62172 | .62552 | .62930 | .63307 | .63683 | .64058 | .64431 | .64803 | .65173 |
| +0.4     | .65542 | .65910 | .66276 | .66640 | .67003 | .67364 | .67724 | .68082 | .68439 | .68793 |
| +0.5     | .69146 | .69497 | .69847 | .70194 | .70540 | .70884 | .71226 | .71566 | .71904 | .72240 |
| +0.6     | .72575 | .72907 | .73237 | .73565 | .73891 | .74215 | .74537 | .74857 | .75175 | .75490 |
| +0.7     | .75804 | .76115 | .76424 | .76730 | .77035 | .77337 | .77637 | .77935 | .78230 | .78524 |
| +0.8     | .78814 | .79103 | .79389 | .79673 | .79955 | .80234 | .80511 | .80785 | .81057 | .81327 |
| +0.9     | .81594 | .81859 | .82121 | .82381 | .82639 | .82894 | .83147 | .83398 | .83646 | .83891 |
| +1       | .84134 | .84375 | .84614 | .84849 | .85083 | .85314 | .85543 | .85769 | .85993 | .86214 |
| +1.1     | .86433 | .86650 | .86864 | .87076 | .87286 | .87493 | .87698 | .87900 | .88100 | .88298 |
| +1.2     | .88493 | .88686 | .88877 | .89065 | .89251 | .89435 | .89617 | .89796 | .89973 | .90147 |
| +1.3     | .90320 | .90490 | .90658 | .90824 | .90988 | .91149 | .91308 | .91466 | .91621 | .91774 |
| +1.4     | .91924 | .92073 | .92220 | .92364 | .92507 | .92647 | .92785 | .92922 | .93056 | .93189 |
| +1.5     | .93319 | .93448 | .93574 | .93699 | .93822 | .93943 | .94062 | .94179 | .94295 | .94408 |
| +1.6     | .94520 | .94630 | .94738 | .94845 | .94950 | .95053 | .95154 | .95254 | .95352 | .95449 |
| +1.7     | .95543 | .95637 | .95728 | .95818 | .95907 | .95994 | .96080 | .96164 | .96246 | .96327 |
| +1.8     | .96407 | .96485 | .96562 | .96638 | .96712 | .96784 | .96856 | .96926 | .96995 | .97062 |
| +1.9     | .97128 | .97193 | .97257 | .97320 | .97381 | .97441 | .97500 | .97558 | .97615 | .97670 |
| +2       | .97725 | .97778 | .97831 | .97882 | .97932 | .97982 | .98030 | .98077 | .98124 | .98169 |
| +2.1     | .98214 | .98257 | .98300 | .98341 | .98382 | .98422 | .98461 | .98500 | .98537 | .98574 |
| +2.2     | .98610 | .98645 | .98679 | .98713 | .98745 | .98778 | .98809 | .98840 | .98870 | .98899 |
| +2.3     | .98928 | .98956 | .98983 | .99010 | .99036 | .99061 | .99086 | .99111 | .99134 | .99158 |
| +2.4     | .99180 | .99202 | .99224 | .99245 | .99266 | .99286 | .99305 | .99324 | .99343 | .99361 |
| +2.5     | .99379 | .99396 | .99413 | .99430 | .99446 | .99461 | .99477 | .99492 | .99506 | .99520 |
| +2.6     | .99534 | .99547 | .99560 | .99573 | .99585 | .99598 | .99609 | .99621 | .99632 | .99643 |
| +2.7     | .99653 | .99664 | .99674 | .99683 | .99693 | .99702 | .99711 | .99720 | .99728 | .99736 |
| +2.8     | .99744 | .99752 | .99760 | .99767 | .99774 | .99781 | .99788 | .99795 | .99801 | .99807 |
| +2.9     | .99813 | .99819 | .99825 | .99831 | .99836 | .99841 | .99846 | .99851 | .99856 | .99861 |
| +3       | .99865 | .99869 | .99874 | .99878 | .99882 | .99886 | .99889 | .99893 | .99896 | .99900 |

## standardization of a normal variable



$$x \approx N_{\mu, \sigma}(x):$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

standardization

$$\Rightarrow Z = \frac{x - \mu}{\sigma} \Rightarrow$$

$$Z \approx N(0, 1):$$

$$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}Z^2}$$

it can be shown that

$$N_{\mu, \sigma}(x) = \frac{1}{\sigma} N\left(\frac{x - \mu}{\sigma}\right)$$

and

$$\Phi_{\mu, \sigma}(x) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$



## CENTRAL LIMIT THEOREM (simplified)

given  $n$  random variables **independent** and **identically distributed**  $X_1, X_2, \dots, X_n$ , their **arithmetic mean** tends to be normally distributed:

$$\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n} \approx N_{\mu, \sigma}(x) \quad \text{as } n \rightarrow \infty$$

also their **sum**:

$$S_n = X_1 + X_2 + \dots + X_n \approx N_{\mu, \sigma}(x) \quad \text{as } n \rightarrow \infty$$



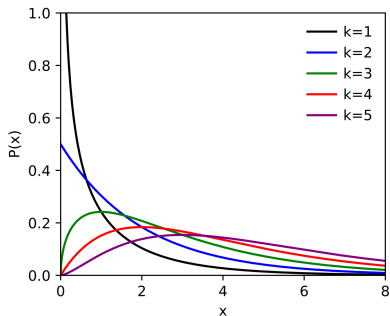
## $\chi_k^2$ DISTRIBUTION

- $\chi_k^2 = \sum_{i=1,k} x_i^2 = x_1^2 + x_2^2 + \dots + x_k^2$

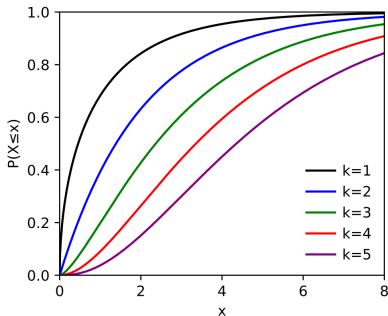
with  $x_i \approx N_{0,1}(x)$

- $k$ =degrees of freedom

density:



cumulative:



# Sampling

inference based on **population** and **samples**

- **population**: large (infinite) collection of units on which a variable can be measured
- **sample** (with or without repetition): finite subset of units of the population
- **random sample**: a sample extracted randomly from the population

example:

- the mean of the age of a population of 1000 units is unknown
- a random sample of 100 units is extracted and its **sample mean** is computed

the **sample mean** is a random variable!

⇒ it must have a probability distribution (e.g. a mean and a variance)...

# Inference

what is **inference**? two settings...

- **interval estimate:**

- I know a random variable distribution on a population
- I observe the random variable on a sample of the population

can I estimate an interval in which a certain parameter (e.g.  $\mu$ ) of the distribution of the population may lie with high probability?

- **hypothesis rejection:**

- I know a random variable distribution on a population
- I make an hypothesis on the value of a certain parameter of the distribution
- I observe the random variable on a sample of the population

do the observations on the sample confirm my hypothesis?

## confidence interval for the mean

- a random variable  $X$  in a population is distributed as  $N_{\mu,\sigma}(x)$  with  $\sigma$  known and  $\mu$  **unknown**
- a random sample  $S := \{x_1, x_2, \dots, x_n\}$  is observed
- **sample mean**  $\bar{x} = \frac{1}{n} \sum_{i=1,n} x_i$  is a random variable

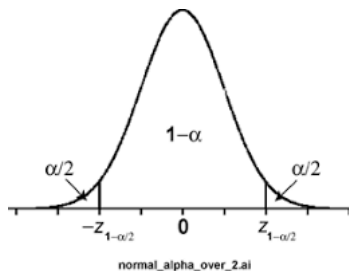
it can be shown that  $\bar{x} \approx N_{\mu, \frac{\sigma}{\sqrt{n}}}(\bar{x})$ , i.e.  $E(\bar{x}) = \mu$ ,  $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

$$\Rightarrow N_{\mu, \frac{\sigma}{\sqrt{n}}} \left( \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \right) \approx N(Z)$$

we want determine an interval of values in which the unknown  $\mu$  falls with probability  $1 - \alpha$  (e.g. 95% if  $\alpha = 0.05$ )...

exploit the fact that  $\frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$  is a standard normal variable!

## confidence interval for the mean



$$P\left\{Z_{\frac{\alpha}{2}} \leq \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq Z_{1-\frac{\alpha}{2}}\right\} = 1 - \alpha$$

$$P\left\{-\bar{x} + \frac{\sigma}{\sqrt{n}}Z_{\frac{\alpha}{2}} \leq -\mu \leq -\bar{x} + \frac{\sigma}{\sqrt{n}}Z_{1-\frac{\alpha}{2}}\right\} = 1 - \alpha$$

$$P\left\{\bar{x} - \frac{\sigma}{\sqrt{n}}Z_{1-\frac{\alpha}{2}} \leq \mu \leq \bar{x} - \frac{\sigma}{\sqrt{n}}Z_{\frac{\alpha}{2}}\right\} = 1 - \alpha$$

$$P\left\{\bar{x} - \frac{\sigma}{\sqrt{n}}Z_{1-\frac{\alpha}{2}} \leq \mu \leq \bar{x} + \frac{\sigma}{\sqrt{n}}Z_{1-\frac{\alpha}{2}}\right\} = 1 - \alpha$$

example:

the viral load of asymptomatic covid patients in L'Aquila is distributed as a normal random variable with standard deviation 0.7 and **unknown mean**. The viral load is observed on a sample of 25 asymptomatic patients and the sample mean is 5.23 copies/ml (in Log10 scale). Define an interval of values  $[a, b]$  in which the mean of the whole population lies with a 95% probability.

$$\text{DATA: } \begin{cases} \sigma = 0.7 \\ \bar{x} = 5.23 \\ n = 25 \\ 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 \end{cases}$$

$$\text{UNKNOWN: } \begin{cases} a \\ b \end{cases}$$

$$P\left\{5.23 - \frac{0.7}{\sqrt{25}} Z_{0.975} \leq \mu \leq 5.23 + \frac{0.7}{\sqrt{25}} Z_{0.975}\right\} = 0.95$$

| z    | 0     | 0.01  | 0.02  | 0.03  | 0.04  | 0.05  | 0.06  | 0.07  |
|------|-------|-------|-------|-------|-------|-------|-------|-------|
| +0   | .5000 | .5039 | .5078 | .5117 | .5155 | .5194 | .5232 | .5270 |
| +0.1 | .5398 | .5438 | .5477 | .5517 | .5556 | .5596 | .5636 | .5674 |
| +0.2 | .5792 | .5831 | .5870 | .5909 | .5948 | .5987 | .6025 | .6064 |
| +0.3 | .6179 | .6217 | .6255 | .6293 | .6330 | .6368 | .6405 | .6443 |
| +0.4 | .6542 | .6580 | .6617 | .6654 | .6691 | .6728 | .6764 | .6801 |
| +0.5 | .6914 | .6949 | .6984 | .7019 | .7054 | .7088 | .7122 | .7156 |
| +0.6 | .7257 | .7291 | .7324 | .7356 | .7389 | .7421 | .7453 | .7485 |
| +0.7 | .7580 | .7611 | .7642 | .7673 | .7703 | .7733 | .7763 | .7793 |
| +0.8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8078 |
| +0.9 | .8159 | .8185 | .8211 | .8238 | .8263 | .8289 | .8314 | .8338 |
| +1   | .8413 | .8437 | .8461 | .8484 | .8508 | .8531 | .8554 | .8576 |
| +1.1 | .8643 | .8665 | .8686 | .8707 | .8728 | .8749 | .8769 | .8790 |
| +1.2 | .8849 | .8868 | .8887 | .8906 | .8925 | .8943 | .8961 | .8979 |
| +1.3 | .9032 | .9049 | .9065 | .9082 | .9098 | .9114 | .9130 | .9146 |
| +1.4 | .9192 | .9207 | .9222 | .9236 | .9250 | .9264 | .9278 | .9292 |
| +1.5 | .9331 | .9344 | .9357 | .9369 | .9382 | .9394 | .9406 | .9419 |
| +1.6 | .9450 | .9463 | .9478 | .9484 | .9495 | .9505 | .9514 | .9524 |
| +1.7 | .9554 | .9563 | .9572 | .9581 | .9590 | .9599 | .9608 | .9616 |
| +1.8 | .9640 | .9648 | .9656 | .9663 | .9671 | .9678 | .9685 | .9692 |
| +1.9 | .9712 | .9719 | .9727 | .9732 | .9738 | .9741 | .9750 | .9758 |
| +2   | .9772 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9807 |

$$Z_{0.975} = 1.96, \quad \frac{0.7}{\sqrt{25}} = 0.14$$

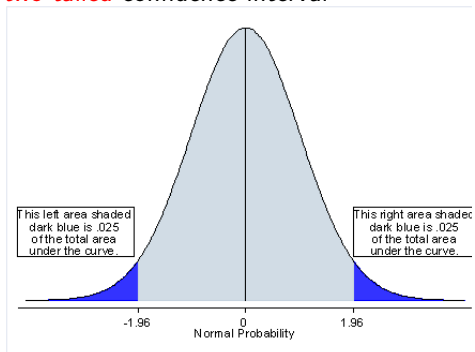
$$a = 5.23 - 1.96 * 0.14 = 4.9556$$

$$b = 5.23 + 1.96 * 0.14 = 5.5044$$

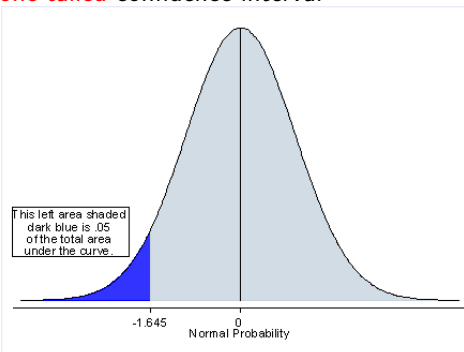
$$P\{4.9556 \leq \mu \leq 5.5044\} = 0.95$$



## two-tailed confidence interval



## one-tailed confidence interval



what is the value  $a$  such the mean of the viral load is greater than or equal to  $a$  ( $a \leq \mu$ ) with probability 95%?

$$P\left\{5.23 - \frac{0.7}{\sqrt{25}} Z_{0.95} \leq \mu\right\} = 0.95$$

$$Z_{0.95} = 1.645, \quad \frac{0.7}{\sqrt{25}} = 0.14, \quad a = 5.23 - 1.645 * 0.14 = 4.9997$$

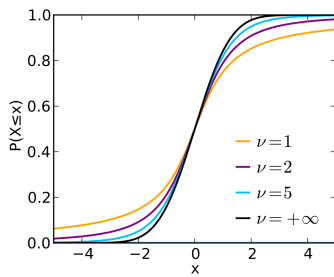
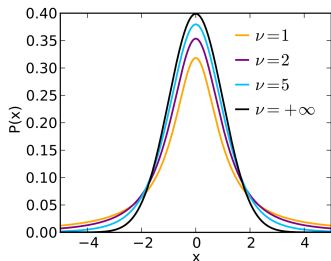
$$P\{4.9997 \leq \mu\} = 0.95$$



## confidence interval for the mean

what if  $\sigma$  is unknown?  $\Rightarrow$  STUDENT'S-t DISTRIBUTION

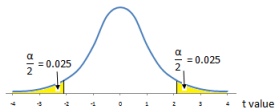
- $t_n = \frac{Z}{\frac{V}{\sqrt{n}}}$  with  $\begin{cases} Z \approx N(x) \\ V^2 \approx \chi_n^2 \end{cases}$ , and  $\nu = n$  are the degrees of freedom
- symmetric around the mean 0 like  $N(x)$
- "shorter" and "fatter" than  $N(x)$
- tends to  $N(x)$  when  $\nu$  grows
- table of values available for the cumulative distribution





## Student's t Distribution Table

For example, the t value for  
18 degrees of freedom  
is 2.101 for 95% confidence  
interval (**2-Tail**  $\alpha = 0.05$ ).



|    | 90%    | 95%    | 97.5%   | 99%     | 99.5%   | 99.95%   | 1-Tail Confidence Level |
|----|--------|--------|---------|---------|---------|----------|-------------------------|
|    | 80%    | 90%    | 95%     | 98%     | 99%     | 99.9%    | 2-Tail Confidence Level |
|    | 0.100  | 0.050  | 0.025   | 0.010   | 0.005   | 0.0005   | 1-Tail Alpha            |
| df | 0.20   | 0.10   | 0.05    | 0.02    | 0.01    | 0.001    | 2-Tail Alpha            |
| 1  | 3.0777 | 6.3138 | 12.7062 | 31.8205 | 63.6567 | 636.6192 |                         |
| 2  | 1.8856 | 2.9200 | 4.3027  | 6.9646  | 9.9248  | 31.5991  |                         |
| 3  | 1.6377 | 2.3534 | 3.1824  | 4.5407  | 5.8409  | 12.9240  |                         |
| 4  | 1.5332 | 2.1318 | 2.7764  | 3.7469  | 4.6041  | 8.6103   |                         |
| 5  | 1.4759 | 2.0150 | 2.5706  | 3.3649  | 4.0321  | 6.8688   |                         |
| 6  | 1.4398 | 1.9432 | 2.4469  | 3.1427  | 3.7074  | 5.9588   |                         |
| 7  | 1.4149 | 1.8946 | 2.3646  | 2.9980  | 3.4995  | 5.4079   |                         |
| 8  | 1.3968 | 1.8595 | 2.3060  | 2.8965  | 3.3554  | 5.0413   |                         |
| 9  | 1.3830 | 1.8331 | 2.2622  | 2.8214  | 3.2498  | 4.7809   |                         |
| 10 | 1.3722 | 1.8125 | 2.2281  | 2.7638  | 3.1693  | 4.5869   |                         |
| 11 | 1.3634 | 1.7959 | 2.2010  | 2.7181  | 3.1058  | 4.4370   |                         |
| 12 | 1.3562 | 1.7823 | 2.1788  | 2.6810  | 3.0545  | 4.3178   |                         |
| 13 | 1.3502 | 1.7709 | 2.1604  | 2.6503  | 3.0123  | 4.2208   |                         |
| 14 | 1.3450 | 1.7613 | 2.1448  | 2.6245  | 2.9768  | 4.1405   |                         |
| 15 | 1.3406 | 1.7531 | 2.1314  | 2.6025  | 2.9467  | 4.0728   |                         |
| 16 | 1.3368 | 1.7459 | 2.1199  | 2.5835  | 2.9208  | 4.0150   |                         |
| 17 | 1.3334 | 1.7396 | 2.1098  | 2.5669  | 2.8982  | 3.9651   |                         |
| 18 | 1.3304 | 1.7341 | 2.1009  | 2.5524  | 2.8784  | 3.9216   |                         |
| 19 | 1.3277 | 1.7291 | 2.0930  | 2.5395  | 2.8609  | 3.8834   |                         |
| 20 | 1.3253 | 1.7247 | 2.0860  | 2.5280  | 2.8453  | 3.8495   |                         |
| 21 | 1.3232 | 1.7207 | 2.0796  | 2.5176  | 2.8314  | 3.8193   |                         |
| 22 | 1.3212 | 1.7171 | 2.0739  | 2.5083  | 2.8188  | 3.7921   |                         |
| 23 | 1.3195 | 1.7139 | 2.0687  | 2.4999  | 2.8073  | 3.7676   |                         |
| 24 | 1.3178 | 1.7109 | 2.0639  | 2.4922  | 2.7969  | 3.7454   |                         |
| 25 | 1.3163 | 1.7081 | 2.0595  | 2.4851  | 2.7874  | 3.7251   |                         |

## confidence interval for the mean

- a random variable  $X$  in a population is distributed as  $N_{\mu,\sigma}(x)$  with  $\mu$  and  $\sigma$  **unknown**
- a random sample  $S := \{x_1, x_2, \dots, x_n\}$  is observed
- **sample mean**  $\bar{x} = \frac{1}{n} \sum_{i=1,n} x_i$ , **sample standard deviation**  $\sigma_S = \sqrt{\frac{\sum_{i=1,n} (x_i - \bar{x})^2}{n-1}}$

it can be shown that  $\frac{\bar{x} - \mu}{\frac{\sigma_S}{\sqrt{n}}} \approx t_{n-1}$ , i.e. a Student's-t distribution with  $\nu = n - 1$  degrees of freedom

$$P\left\{t_{n-1}^{\frac{\alpha}{2}} \leq \frac{\bar{x} - \mu}{\frac{\sigma_S}{\sqrt{n}}} \leq t_{n-1}^{1-\frac{\alpha}{2}}\right\} = 1 - \alpha$$

$$P\left\{\bar{x} - \frac{\sigma_S}{\sqrt{n}} t_{n-1}^{1-\frac{\alpha}{2}} \leq \mu \leq \bar{x} + \frac{\sigma_S}{\sqrt{n}} t_{n-1}^{1-\frac{\alpha}{2}}\right\} = 1 - \alpha$$

$$\text{if } n > 30 \quad P\left\{\bar{x} - \frac{\sigma_S}{\sqrt{n}} Z_{1-\frac{\alpha}{2}} \leq \mu \leq \bar{x} + \frac{\sigma_S}{\sqrt{n}} Z_{1-\frac{\alpha}{2}}\right\} = 1 - \alpha$$

example 2.0:

consider the viral load of asymptomatic covid patients in L'Aquila is distributed as a normal random variable with **unknown mean and standard deviation**. The viral load is observed on a sample of 25 asymptomatic patients and the sample mean is 5.23 copies/ml while the sample standard deviation is 0.7. Define an interval of values  $[a, b]$  in which the mean of the whole population lies with a 95% probability.

$$\text{DATA: } \begin{cases} \sigma_S = 0.7 \\ \bar{x} = 5.23 \\ n = 25 \\ 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 \end{cases}$$

$$\text{UNKNOWN: } \begin{cases} a \\ b \end{cases}$$

$$P\left\{5.23 - \frac{0.7}{\sqrt{25}} t_{24}^{0.975} \leq \mu \leq 5.23 + \frac{0.7}{\sqrt{25}} t_{24}^{0.975}\right\} = 0.95$$

|    |        |        |        |
|----|--------|--------|--------|
| 16 | 1.3368 | 1.7459 | 2.1199 |
| 17 | 1.3334 | 1.7396 | 2.1098 |
| 18 | 1.3304 | 1.7341 | 2.1009 |
| 19 | 1.3277 | 1.7291 | 2.0930 |
| 20 | 1.3253 | 1.7247 | 2.0860 |
| 21 | 1.3232 | 1.7207 | 2.0796 |
| 22 | 1.3212 | 1.7171 | 2.0739 |
| 23 | 1.3195 | 1.7139 | 2.0687 |
| 24 | 1.3178 | 1.7109 | 2.0639 |
| 25 | 1.3163 | 1.7081 | 2.0595 |

$$t_{24}^{0.975} = 2.0639, \quad \frac{0.7}{\sqrt{25}} = 0.14$$

$$a = 5.23 - 2.0639 * 0.14 = 4.941$$

$$b = 5.23 + 2.0639 * 0.14 = 5.519$$

$$P\{4.941 \leq \mu \leq 5.519\} = 0.95$$



## test of hypothesis

two alternative hypotheses

- $H_0$  (null hypothesis): a random variable follows a **certain** distribution with **certain** parameters
- $H_1$ :  $H_0$  is false (different alternatives are possible)

by observing a sample of the random variable is it possible to say if the random variable follows the distribution of  $H_0$  with a certain confidence level  $1 - \alpha$ ?

- 1 not possible to reject  $H_0$  (different from accepting  $H_0$ )
- 2  $H_0$  is rejected ( $H_0$  is false and  $H_1$  is true)

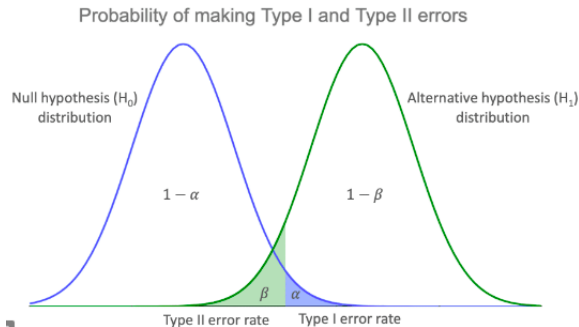
two settings

- one sample test
- two samples test

## test of hypothesis

two possible errors

- **type 1** error: rejecting  $H_0$  when  $H_0$  is true (probability  $\alpha$ )
- **type 2** error: not rejecting  $H_0$  when  $H_0$  is false (probability  $\beta$ )



$1 - \beta$  power of the test: probability of rejecting  $H_0$  when  $H_0$  is false

## one sample hypothesis test

- a random variable  $X$  in a population is normally distributed with known standard deviation  $\sigma$  and an hypothesis test is made on the value of its mean  $\mu$  with confidence level  $1 - \alpha$ 
  - ▶  $H_0: \mu = \mu_0$
  - ▶  $H_1: \mu > \mu_0$  (one sided alternative)
- a random sample  $S := \{x_1, x_2, \dots, x_n\}$  is observed with sample mean  $\bar{x} = \frac{1}{n} \sum_{i=1, n} x_i$

if  $H_0$  is "true"  $\frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \approx N(x)$  then

$$P\left\{\frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \leq Z_{1-\alpha}\right\} = 1 - \alpha \quad \Rightarrow \quad P\{\bar{x} \leq \mu_0 + \frac{\sigma}{\sqrt{n}} Z_{1-\alpha}\} = 1 - \alpha$$

if  $\begin{cases} \bar{x} \leq \mu_0 + \frac{\sigma}{\sqrt{n}} Z_{1-\alpha}, & H_0 \text{ not rejected (error type 2 with probability } \beta) \\ \bar{x} > \mu_0 + \frac{\sigma}{\sqrt{n}} Z_{1-\alpha}, & H_0 \text{ rejected (error type 1 with probability } \alpha) \end{cases}$



example:

a normal random variable has standard deviation  $\sigma = 3.6$  with unknown mean. Consider the following hypotheses

- $H_0: \mu = 12$
- $H_1: \mu > 12$

by observing a sample of 16 realizations of the random variable with sample mean 13.7, can we accept  $H_0$  with confidence level 95% ( $\alpha = 0.05$ )?

| $z$  | 0      | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   |
|------|--------|--------|--------|--------|--------|--------|--------|--------|
| +0   | .50000 | .50399 | .50798 | .51197 | .51595 | .51994 | .52392 | .52790 |
| +0.1 | .53983 | .54380 | .54776 | .55172 | .55567 | .55966 | .56360 | .56749 |
| +0.2 | .57926 | .58317 | .58706 | .59095 | .59483 | .59871 | .60257 | .60642 |
| +0.3 | .61791 | .62172 | .62552 | .62930 | .63307 | .63683 | .64058 | .64431 |
| +0.4 | .65542 | .65910 | .66276 | .66640 | .67003 | .67364 | .67724 | .68082 |
| +0.5 | .69146 | .69497 | .69847 | .70194 | .70540 | .70884 | .71226 | .71566 |
| +0.6 | .72575 | .72907 | .73237 | .73565 | .73891 | .74215 | .74537 | .74857 |
| +0.7 | .75804 | .76115 | .76424 | .76730 | .77035 | .77337 | .77637 | .77935 |
| +0.8 | .78814 | .79103 | .79389 | .79673 | .79955 | .80234 | .80511 | .80785 |
| +0.9 | .81594 | .81859 | .82121 | .82381 | .82639 | .82894 | .83147 | .83398 |
| +1   | .84134 | .84375 | .84614 | .84849 | .85083 | .85314 | .85543 | .85769 |
| +1.1 | .86433 | .86650 | .86864 | .87076 | .87286 | .87493 | .87698 | .87900 |
| +1.2 | .88493 | .88686 | .88877 | .89065 | .89251 | .89435 | .89617 | .89796 |
| +1.3 | .90320 | .90490 | .90658 | .90824 | .90988 | .91149 | .91308 | .91466 |
| +1.4 | .91924 | .92073 | .92220 | .92364 | .92507 | .92647 | .92785 | .92922 |
| +1.5 | .93319 | .93448 | .93574 | .93699 | .93822 | .93943 | .94062 | .94179 |
| +1.6 | .94520 | .94630 | .94738 | .94845 | .94950 | .95053 | .95154 | .95254 |
| +1.7 | .95543 | .95637 | .95728 | .95818 | .95907 | .95994 | .96080 | .96164 |
| +1.8 | .96407 | .96485 | .96562 | .96638 | .96712 | .96784 | .96856 | .96926 |
| +1.9 | .97128 | .97193 | .97257 | .97320 | .97381 | .97441 | .97500 | .97558 |
| +2   | .97725 | .97778 | .97831 | .97882 | .97932 | .97982 | .98030 | .98077 |

$$12 + \frac{3.6}{\sqrt{16}} Z_{0.95} = 12 + 0.9 * 1.645 =$$

$$= 12 + 1.4805 = 13.4805$$

$$13.7 > 13.4805 \Rightarrow H_0 \text{ rejected}$$

## two samples hypothesis test

- two random variables  $X$  and  $Y$  are normally distributed with the same variance and an hypothesis test is made on their means  $\mu_X, \mu_Y$  with confidence level  $1 - \alpha$ 
  - ▶  $H_0: \mu_X = \mu_Y$
  - ▶  $H_1: \mu_X \neq \mu_Y$  (two sided alternative)
- random samples  $S_X := \{x_1, x_2, \dots, x_{n_X}\}, S_Y := \{y_1, y_2, \dots, y_{n_Y}\}$  are observed with sample means  $\bar{x}, \bar{y}$  and sample variances  $\sigma_X, \sigma_Y$

it is possible to show that if  $H_0$  is "true" ( $\mu_X = \mu_Y$ ) then

$$\Psi = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_X^2(n_X-1) + \sigma_Y^2(n_Y-1)}{n_X + n_Y - 2} \left( \frac{1}{n_X} + \frac{1}{n_Y} \right)}} \approx t_{n_X + n_Y - 2}$$

$$P\left\{t_{n_X + n_Y - 2}^{\frac{\alpha}{2}} \leq \Psi \leq t_{n_X + n_Y - 2}^{1 - \frac{\alpha}{2}}\right\} = 1 - \alpha$$

$$\text{if } \begin{cases} t_{n_X + n_Y - 2}^{\frac{\alpha}{2}} \leq \Psi \leq t_{n_X + n_Y - 2}^{1 - \frac{\alpha}{2}}, & H_0 \text{ not rejected} \\ \Psi < t_{n_X + n_Y - 2}^{\frac{\alpha}{2}} \text{ or } \Psi > t_{n_X + n_Y - 2}^{1 - \frac{\alpha}{2}}, & H_0 \text{ rejected} \end{cases}$$



example:

two samples  $X$  and  $Y$  normally distributed with the same variance are observed...

$$\text{DATA: } \begin{cases} \bar{x} = 26.8 \\ \sigma_X = 1.7 \\ n_X = 30 \\ \bar{y} = 27.2 \\ \sigma_Y = 0.8 \\ n_Y = 30 \end{cases}$$

$$\text{HYPOTHESES: } \begin{cases} H_0 : \mu_X = \mu_Y \\ H_1 : \mu_X \neq \mu_Y \\ 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 \end{cases}$$

$$\frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_X^2(n_X-1) + \sigma_Y^2(n_Y-1)}{n_X + n_Y - 2} \left( \frac{1}{n_X} + \frac{1}{n_Y} \right)}} = \frac{26.8 - 27.2}{\sqrt{\frac{1.7^2(29) + 0.8^2(29)}{58} \left( \frac{1}{30} + \frac{1}{30} \right)}} = -1.166$$

$$t_{58}^{0.025} \dots$$

$$t_{58}^{0.975} \dots$$

example:

two samples  $X$  and  $Y$  normally distributed with the same variance are observed...

$$\text{DATA: } \begin{cases} \bar{x} = 26.8 \\ \sigma_X = 1.7 \\ n_X = 30 \\ \bar{y} = 27.2 \\ \sigma_Y = 0.8 \\ n_Y = 30 \end{cases} \quad \text{HYPOTHESES: } \begin{cases} H_0 : \mu_X = \mu_Y \\ H_1 : \mu_X \neq \mu_Y \\ 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 \end{cases}$$

$$\frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_X^2(n_X-1) + \sigma_Y^2(n_Y-1)}{n_X + n_Y - 2} \left( \frac{1}{n_X} + \frac{1}{n_Y} \right)}} = \frac{26.8 - 27.2}{\sqrt{\frac{1.7^2(29) + 0.8^2(29)}{58} \left( \frac{1}{30} + \frac{1}{30} \right)}} = -1.166$$

$$t_{58}^{0.025} \approx Z_{0.025} = -1.96$$

$$t_{58}^{0.975} \approx Z_{0.975} = 1.96$$

$$-1.96 < -1.166 < 1.96 \Rightarrow H_0 \text{ not rejected}$$

$\chi^2$  test is used for categorical random variables

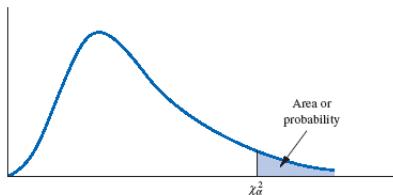
- a random variable  $E$  can assume a finite set of  $k$  values  $E_1, E_2, \dots, E_k$
- according to its probability distribution the expected frequencies of the values are  $f_1, f_2, \dots, f_k$  with  $f_i \in (0, 1)$
- frequencies  $o_1, o_2, \dots, o_k$  are observed on a random sample of a random variable  $X$
- $H_0: X = E$   
 $H_1: X \neq E$

it is possible to show that if  $H_0$  is true, then

$$\sum_{i=1,k} \frac{(o_i - f_i)^2}{f_i} \approx \chi_{k-1}^2$$

$\Rightarrow \chi^2$  table can be used to verify  $H_0$

TABLE 11.1 SELECTED VALUES FROM THE CHI-SQUARE DISTRIBUTION TABLE\*



| Degrees of Freedom | Area in Upper Tail |        |        |        |        |        |        |        |
|--------------------|--------------------|--------|--------|--------|--------|--------|--------|--------|
|                    | .99                | .975   | .95    | .90    | .10    | .05    | .025   | .01    |
| 1                  | .000               | .001   | .004   | .016   | 2.706  | 3.841  | 5.024  | 6.635  |
| 2                  | .020               | .051   | .103   | .211   | 4.605  | 5.991  | 7.378  | 9.210  |
| 3                  | .115               | .216   | .352   | .584   | 6.251  | 7.815  | 9.348  | 11.345 |
| 4                  | .297               | .484   | .711   | 1.064  | 7.779  | 9.488  | 11.143 | 13.277 |
| 5                  | .554               | .831   | 1.145  | 1.610  | 9.236  | 11.070 | 12.832 | 15.086 |
| 6                  | .872               | 1.237  | 1.635  | 2.204  | 10.645 | 12.592 | 14.449 | 16.812 |
| 7                  | 1.239              | 1.690  | 2.167  | 2.833  | 12.017 | 14.067 | 16.013 | 18.475 |
| 8                  | 1.647              | 2.180  | 2.733  | 3.490  | 13.362 | 15.507 | 17.535 | 20.090 |
| 9                  | 2.088              | 2.700  | 3.325  | 4.168  | 14.684 | 16.919 | 19.023 | 21.666 |
| 10                 | 2.558              | 3.247  | 3.940  | 4.865  | 15.987 | 18.307 | 20.483 | 23.209 |
| 11                 | 3.053              | 3.816  | 4.575  | 5.578  | 17.275 | 19.675 | 21.920 | 24.725 |
| 12                 | 3.571              | 4.404  | 5.226  | 6.304  | 18.549 | 21.026 | 23.337 | 26.217 |
| 13                 | 4.107              | 5.009  | 5.892  | 7.041  | 19.812 | 22.362 | 24.736 | 27.688 |
| 14                 | 4.660              | 5.629  | 6.571  | 7.790  | 21.064 | 23.685 | 26.119 | 29.141 |
| 15                 | 5.229              | 6.262  | 7.261  | 8.547  | 22.307 | 24.996 | 27.488 | 30.578 |
| 16                 | 5.812              | 6.908  | 7.962  | 9.312  | 23.542 | 26.296 | 28.845 | 32.000 |
| 17                 | 6.408              | 7.564  | 8.672  | 10.085 | 24.769 | 27.587 | 30.191 | 33.409 |
| 18                 | 7.015              | 8.231  | 9.390  | 10.865 | 25.989 | 28.869 | 31.526 | 34.805 |
| 19                 | 7.633              | 8.907  | 10.117 | 11.651 | 27.204 | 30.144 | 32.852 | 36.191 |
| 20                 | 8.260              | 9.591  | 10.851 | 12.443 | 28.412 | 31.410 | 34.170 | 37.566 |
| 21                 | 8.897              | 10.283 | 11.591 | 13.240 | 29.615 | 32.671 | 35.479 | 38.932 |
| 22                 | 9.542              | 10.982 | 12.338 | 14.041 | 30.813 | 33.924 | 36.781 | 40.289 |
| 23                 | 10.196             | 11.689 | 13.091 | 14.848 | 32.007 | 35.172 | 38.076 | 41.638 |
| 24                 | 10.856             | 12.401 | 13.848 | 15.659 | 33.196 | 36.415 | 39.364 | 42.980 |
| 25                 | 11.524             | 13.121 | 14.611 | 16.473 | 34.382 | 37.652 | 40.646 | 44.314 |

example:

verify if a 6-sided dice is unbiased having observed the outcomes of 600 throws with confidence level 99%.

|                 | 1   | 2   | 3   | 4   | 5   | 6   |
|-----------------|-----|-----|-----|-----|-----|-----|
| <i>observed</i> | 120 | 87  | 103 | 100 | 104 | 86  |
| <i>expected</i> | 100 | 100 | 100 | 100 | 100 | 100 |

HYPOTHESES:

$$\begin{cases} H_0 : \text{dice is unbiased} \\ H_1 : \text{dice is biased} \\ 1 - \alpha = 0.99 \Rightarrow \alpha = 0.01 \end{cases}$$

$$\begin{aligned} \sum_{i=1,k} \frac{(o_i - f_i)^2}{f_i} &= \frac{(\frac{120-100}{600})^2}{100/600} + \frac{(\frac{87-100}{600})^2}{100/600} + \frac{(\frac{103-100}{600})^2}{100/600} + \frac{(\frac{100-100}{600})^2}{100/600} + \\ &+ \frac{(\frac{104-100}{600})^2}{100/600} + \frac{(\frac{86-100}{600})^2}{100/600} = 0.013167 \end{aligned}$$

$$\chi_5^{20.01} = 15.086 > 0.013167 \Rightarrow H_0 \text{ not rejected}$$

practical limitations...

- often we don't know the distribution of the random variable
- often samples are not drawn at random

with time and computational power, **empirical distributions** can be built by **resampling** data, and used for

- confidence intervals
- hypothesis testing
- ...

## bootstrap

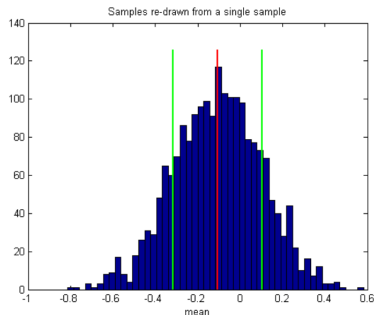
- we want to estimate the mean of a random variable with **unknown distribution**
- a  $n$ -dimensional sample  $S := \{x_1, x_2, \dots, x_n\}$  is observed
- many  $m$ -dimensional ( $m \leq n$ ) sub-samples **with repetition** are extracted from  $S$  and the mean is observed on each subsample to create an **empirical distribution** for the mean

$$\{x_2, x_5, x_{36}, x_1\} \rightarrow \mu_1$$

$$\{x_{21}, x_3, x_{12}, x_3\} \rightarrow \mu_2$$

$\vdots$

$$\{x_6, x_8, x_{15}, x_6\} \rightarrow \mu_k$$



- empirical distribution used to estimate confidence intervals or testing hypotheses

## exercise 1.6

## reshuffling

- investigating the correlation of two random variables  $X$  and  $Y$
- Pearson's correlation coefficient  $\rho_{XY} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$
- breaking the association structure between  $X$  and  $Y$  by randomly **reshuffling** many times one of the two samples and obtaining empirical distribution of the correlation to see if it is reliable

|           | height      | weight      |
|-----------|-------------|-------------|
| person 1  | 1.875714232 | 109.8196777 |
| person 2  | 1.747060363 | 73.68895452 |
| person 3  | 1.882396677 | 96.58434842 |
| person 4  | 1.821966851 | 99.89928152 |
| person 5  | 1.774997615 | 93.68280948 |
| person 6  | 1.708226598 | 69.10431871 |
| person 7  | 1.747141064 | 83.50326143 |
| person 8  | 1.736052294 | 76.25888416 |
| person 9  | 1.702281321 | 79.87196594 |
| person 10 | 1.611794947 | 71.00545308 |
| person 11 | 1.80836271  | 84.7186362  |
| person 12 | 1.81967645  | 97.03849095 |
| person 13 | 1.64506476  | 75.87586733 |
| person 14 | 1.75978998  | 86.00856635 |
| person 15 | 1.758790799 | 84.64111228 |
| person 16 | 1.71819874  | 78.17286625 |
| person 17 | 1.839425242 | 88.99694187 |
| person 18 | 1.624947873 | 78.48909547 |
| person 19 | 1.768857521 | 84.43671674 |
| person 20 | 1.725574523 | 82.8216982  |
| corr      |             | 0.818411356 |



| height      | weight       | height      | weight       | height      | weight      |
|-------------|--------------|-------------|--------------|-------------|-------------|
| 1.875714232 | 75.88        | 1.875714232 | 79.87        | 1.875714232 | 109.82      |
| 1.747060363 | 96.58        | 1.747060363 | 83.5         | 1.747060363 | 86.01       |
| 1.882396677 | 84.64        | 1.882396677 | 96.58        | 1.882396677 | 89          |
| 1.821966851 | 79.87        | 1.821966851 | 71.01        | 1.821966851 | 99.9        |
| 1.774997615 | 97.04        | 1.774997615 | 78.17        | 1.774997615 | 76.26       |
| 1.708226598 | 76.26        | 1.708226598 | 73.69        | 1.708226598 | 96.58       |
| 1.747141064 | 69.1         | 1.747141064 | 82.82        | 1.747141064 | 97.04       |
| 1.736052294 | 93.68        | 1.736052294 | 93.68        | 1.736052294 | 84.44       |
| 1.702281321 | 73.69        | 1.702281321 | 75.88        | 1.702281321 | 78.49       |
| 1.611794947 | 84.72        | 1.611794947 | 84.44        | 1.611794947 | 84.64       |
| 1.80836271  | 82.82        | 1.80836271  | 84.72        | 1.80836271  | 71.01       |
| 1.81967645  | 86.01        | 1.81967645  | 78.49        | 1.81967645  | 75.88       |
| 1.64506476  | 89           | 1.64506476  | 86.01        | 1.64506476  | 78.17       |
| 1.75978998  | 83.5         | 1.75978998  | 76.26        | 1.75978998  | 73.69       |
| 1.758790799 | 99.9         | 1.758790799 | 69.1         | 1.758790799 | 93.68       |
| 1.71819874  | 78.17        | 1.71819874  | 109.82       | 1.71819874  | 82.82       |
| 1.839425242 | 78.49        | 1.839425242 | 89           | 1.839425242 | 84.72       |
| 1.624947873 | 84.44        | 1.624947873 | 84.64        | 1.624947873 | 69.1        |
| 1.768857521 | 71.01        | 1.768857521 | 99.9         | 1.768857521 | 83.5        |
| 1.725574523 | 109.82       | 1.725574523 | 97.04        | 1.725574523 | 79.87       |
|             | -0.129865955 |             | -0.039975309 |             | 0.384984573 |

## exercise 1.7



# Overview

- 1 Introduction
- 2 Basic statistics
- 3 Multivariate statistics and Principal Component Analysis**
- 4 Unsupervised learning: Clustering
- 5 Supervised learning
- 6 Overview on Time Series

# Multivariate statistics

some important questions...

- what data is?
- how we look at data?
- how we manipulate data?
- how we find things in data?

different combination of alternatives are possible!

data is composed by

- individuals/observations/units/samples
- features/characteristics/variables/attributes

minimal data requirements

- **distinguishable** individuals
- **measurable** features

many individuals, many features  $\Rightarrow$  **multivariate** statistics

some assumptions...

- features can be measured on a continuous scale
- data is represented in a grid (matrix):  
    rows = individuals  
    columns = features
- matrix element  $(i, j)$  is the measure of feature  $j$  on individual  $i$
- $m$  is the number of rows (individuals)
- $n$  is the number of columns (features)

$$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \dots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}$$

$$x_j = \begin{pmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{mj} \end{pmatrix}$$

$$X = (x_1, x_2, \dots, x_n)$$

...data has been put into a "space" ...



what do you see?



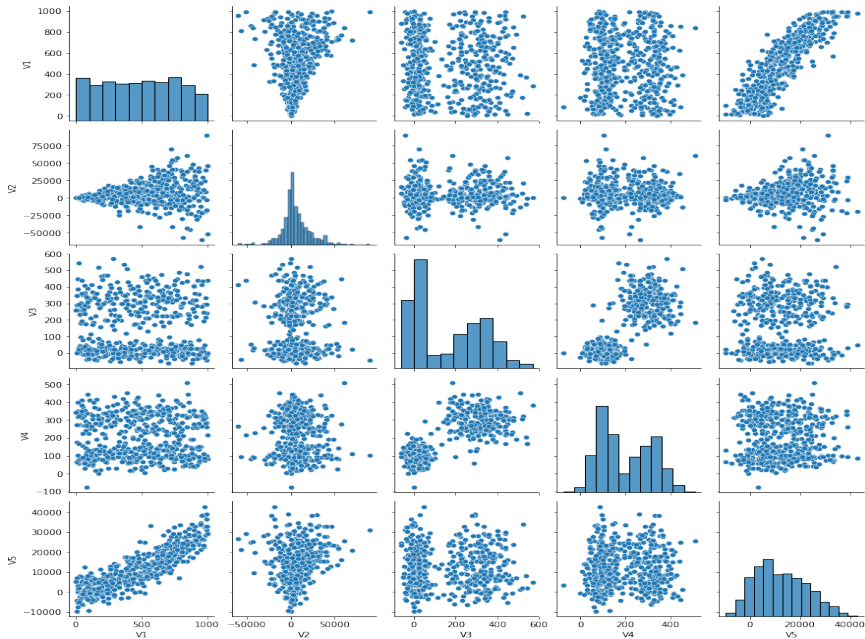
data represented in a **space**  $\Rightarrow$  distance (metric)  $d$

$d : X \times X \rightarrow [0, \infty)$  properties

- ①  $d(x, y) \geq 0$  (non-negativity)
- ②  $d(x, y) = 0 \iff x = y$  (identity of indiscernable)
- ③  $d(x, y) = d(y, x)$  (simmetry)
- ④  $d(x, y) + d(y, z) \geq d(x, z)$  (subadditivity)

exercise 2.1





recall on **correlation**...

a statistical dependence (linear) between two random variables

- Pearson's correlation coefficient:

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_x)(Y - \mu_y)]}{\sigma_X \sigma_Y}$$

- sample version:

$$r_{XY} = \frac{\sum_{i=1,n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1,n} (x_i - \bar{x})^2} \sqrt{\sum_{i=1,n} (y_i - \bar{y})^2}}$$

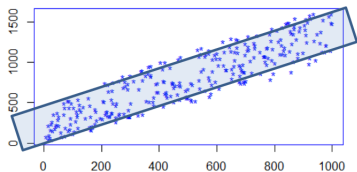
- $\rho_{XY}$  or  $r_{XY} = \begin{cases} 1 & \text{positive correlation} \\ 0 & \text{linear independence} \\ -1 & \text{negative correlation} \end{cases}$

exercise 2.2



beyond correlation, much **information** in data is about **distance** between points  
(the spread!)

- to extract information from data we have to focus on some aspects and discard the others
- data has many variables: is it possible to reduce the dimension of the data space, transform it, and preserve **most of** the distance between points?  
(e.g. humans like to see things in 2 dimensions)
- is it possible to change the system of coordinates to better capture the data variance?



⇒ **orthogonality** is a key aspect

## recall on linear algebra

- 2 vectors are linearly dependent if one can be obtained as a linear combination of the other

$$a = \begin{pmatrix} 2 \\ 1 \\ 4 \\ 3 \end{pmatrix}, b = \begin{pmatrix} 4 \\ 2 \\ 8 \\ 6 \end{pmatrix} \Rightarrow a = 0.5b$$

- $n$  vectors are linearly dependent if any of them can be obtained as a linear combination of the others

$$a = \begin{pmatrix} 2 \\ 1 \\ 4 \\ 3 \end{pmatrix}, b = \begin{pmatrix} 4 \\ 2 \\ 8 \\ 0 \end{pmatrix}, c = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 12 \end{pmatrix} \Rightarrow \begin{cases} a = 0.5b + 0.25c \\ c = 4a - 2b \end{cases}$$

- otherwise they are **linearly independent**

recall on linear algebra

- the **inner product** of 2 vectors  $a = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$ ,  $b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$  can be computed as

$$a^T b = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = \|a\| \|b\| \cos \alpha \quad (\alpha \text{ angle between } a \text{ and } b)$$

$$a = \begin{pmatrix} 2 \\ 1 \\ 4 \\ 3 \end{pmatrix}, b = \begin{pmatrix} 3 \\ 0 \\ 1 \\ 2 \end{pmatrix}, a^T b = (2 \ 1 \ 4 \ 3) \times \begin{pmatrix} 3 \\ 0 \\ 1 \\ 2 \end{pmatrix} = 2 \times 3 + 1 \times 0 + 4 \times 1 + 3 \times 2 = 16$$

- 2 vectors are **orthogonal** if their inner product is zero (they are perpendicular)

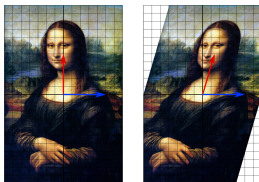
$$a = \begin{pmatrix} 2 \\ 0 \\ -1 \\ 3 \end{pmatrix}, b = \begin{pmatrix} 0 \\ 6 \\ 3 \\ 1 \end{pmatrix}, a^T b = (2 \ 0 \ -1 \ 3) \times \begin{pmatrix} 0 \\ 6 \\ 3 \\ 1 \end{pmatrix} =$$
$$2 \times 0 + 0 \times 6 - 1 \times 3 + 3 \times 1 = 0$$

## recall on linear algebra

- given a matrix  $A^{m \times n}$ , its **rank**  $rk(A)$  is the maximum number of columns or rows linearly independent ( $rk(A) \leq \min\{m, n\}$ )
- given a square matrix  $A^{n \times n}$ , a scalar  $\lambda$  and a vector  $v \in \mathbb{R}^n$  are an **eigenvalue** and an **eigenvector** if they satisfy

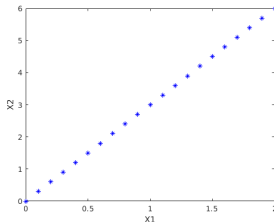
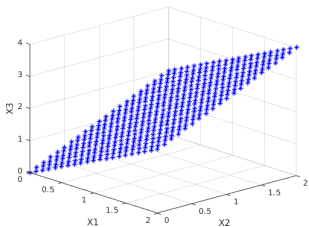
$$Av = \lambda v$$

- ▶ eigenvectors of  $A$  are linearly independent
- ▶ if  $A$  is symmetric there are  $n$  orthogonal eigenvectors (associated to  $n$  eigenvalues)
- ▶ provide an alternative coordinate system for the columns (rows) of  $A$



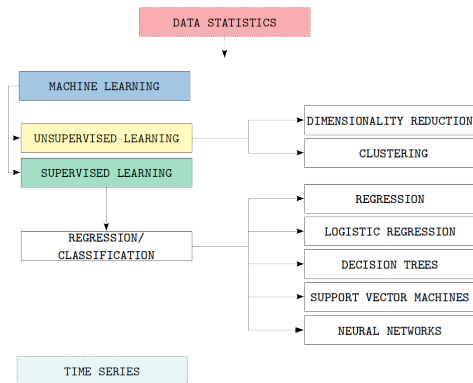
why linear independence, orthogonality, matrix rank, eigenvalues/eigenvectors?  
vectors are measures of variables  $\Rightarrow$  investigating vectors relations provide insights on data (dependence, redundancy,...)

- a set of points with  $n$  coordinates may lie in a space of dimension  $\leq n$   
(points in a plane can lie in a line, points in a 3d-space can lie in a plane...)



- given a set of  $n$  dimensional points one may wonder what is the space of **minimum dimension** where they lie
- in such a space their mutual distance would be preserved
- maybe we are happy with something "less precise", ignoring the less relevant dimensions!

# Principal Component Analysis (PCA)



PCA is a **dimensionality reduction** technique used for

- **exploratory data analysis** (unsupervised learning)
- **feature extraction** (support to supervised learning)

context

- our data are represented with  $X^{m \times n}$

$$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \dots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix} \quad x_i = \begin{pmatrix} x_{1i} \\ x_{2i} \\ \vdots \\ x_{mi} \end{pmatrix} \quad X = (x_1, x_2, \dots, x_n)$$

- data are in a  $n$ -dimensional space with **coordinates**  $x_1, x_2, \dots, x_n$  (features)
- we seek a new system of orthogonal coordinates  $z_1, z_2, \dots, z_p$  (**principal components**) with  $p \leq n$  to represent original data such that
  - it better captures the variance of data  $\Rightarrow$  if  $p < n$  the information loss is "minimal"
  - each new coordinate  $z_i$  is a linear combination of all the original coordinates

$$\begin{aligned} z_1 &= \phi_{11}x_1 + \phi_{21}x_2 + \dots + \phi_{n1}x_n \\ z_2 &= \phi_{12}x_1 + \phi_{22}x_2 + \dots + \phi_{n2}x_n \\ &\vdots \\ z_p &= \phi_{1p}x_1 + \phi_{2p}x_2 + \dots + \phi_{np}x_n \end{aligned} \quad (x_1, x_2, \dots, x_n) \Rightarrow (z_1, z_2, \dots, z_p)$$

- they can be ranked in order of "importance"

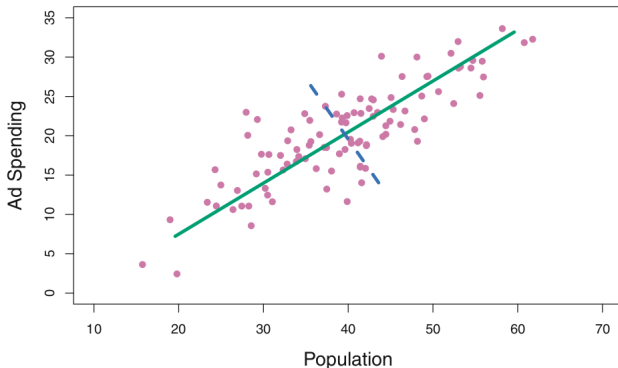


## geometric interpretation

"Principal components are a sequence of projections of the data, mutually uncorrelated and ordered in variance."

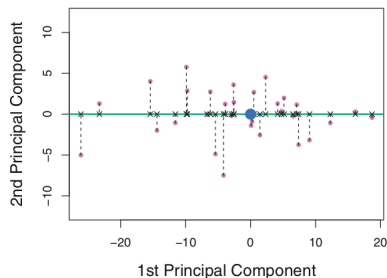
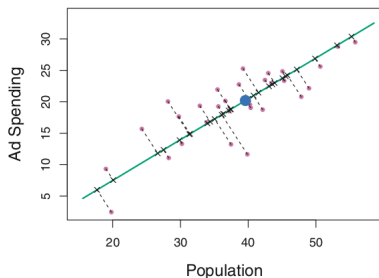
('The Elements of Statistical Learning' page 534)

example: population size (feature 1 or  $x_1$ ) and ad spending (feature 2 or  $x_2$ ) over 100 cities (individuals)





## geometric interpretation



$$z_1 = \phi_{11}x_1 + \phi_{21}x_2$$

$$z_2 = \phi_{12}x_1 + \phi_{22}x_2$$

$$\Rightarrow z_1 = \phi_{11}\text{pop} + \phi_{21}\text{ad}$$

$$z_2 = \phi_{12}\text{pop} + \phi_{22}\text{ad}$$

coordinates  $(z_1, z_2)$  (Principal Component **scores**) determine the position of individuals in the new system!

$\begin{pmatrix} \phi_{11} \\ \phi_{21} \end{pmatrix}, \begin{pmatrix} \phi_{12} \\ \phi_{22} \end{pmatrix}$ : **loadings** vectors associated to 1st and 2nd Principal Components

$$z_1 = 0.839 \times (\text{pop} - \overline{\text{pop}}) + 0.544 \times (\text{ad} - \overline{\text{ad}})$$

## mathematical interpretation

we sequentially determine new coordinates (directions) maximizing the overall variance of data

- center data by columns as  $x_i^{centered} = x_i - \bar{x}_i \Rightarrow \bar{x}_i^{centered} = 0$
- determine the first Principal Component by solving

$$\max_{\phi_{11}, \phi_{21}, \dots, \phi_{n1}} \quad \frac{1}{m} \sum_{i=1, m} \sum_{j=1, n} (\phi_{j1} x_{ij} - \bar{x}_j)^2 = \frac{1}{m} \sum_{i=1, m} \sum_{j=1, n} (\phi_{j1} x_{ij})^2$$

subject to

$$\sum_{j=1, n} \phi_{j1}^2 = 1$$

- determine the second Principal Component by solving

$$\max_{\phi_{12}, \phi_{22}, \dots, \phi_{n2}} \quad \frac{1}{m} \sum_{i=1, m} \sum_{j=1, n} (\phi_{j2} x_{ij})^2$$

subject to

$$\begin{aligned} \sum_{j=1, n} \phi_{j2}^2 &= 1 \\ \sum_{j=1, n} \phi_{j1} \phi_{j2} &= 0 \end{aligned}$$

• ...

## covariance interpretation

- center data by columns as  $x_i^{centered} = x_i - \bar{x}_i \Rightarrow \bar{x}_i^{centered} = 0$
- construct the covariance matrix as

$$C = \begin{pmatrix} \text{COV}(x_1, x_1) & \text{COV}(x_1, x_2) & \dots & \text{COV}(x_1, x_n) \\ \text{COV}(x_2, x_1) & \text{COV}(x_2, x_2) & \dots & \text{COV}(x_2, x_n) \\ \vdots & \vdots & \dots & \vdots \\ \text{COV}(x_n, x_1) & \text{COV}(x_n, x_2) & \dots & \text{COV}(x_n, x_n) \end{pmatrix}$$

C symmetric, positive semidefinite

$$\Rightarrow \begin{cases} n \text{ orthogonal eigenvectors } v_1, v_2, \dots, v_n \\ n \text{ nonnegative eigenvalues } \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0 \end{cases}$$

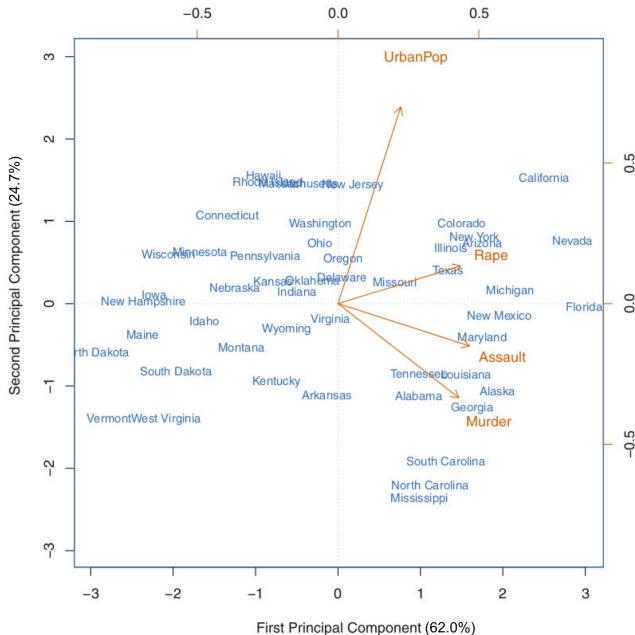
- eigenvectors represent the Principal Components

$$v_1 = \begin{pmatrix} \phi_{11} \\ \phi_{21} \\ \vdots \\ \phi_{n1} \end{pmatrix}, v_2 = \begin{pmatrix} \phi_{12} \\ \phi_{22} \\ \vdots \\ \phi_{n2} \end{pmatrix}, \dots, v_n = \begin{pmatrix} \phi_{1n} \\ \phi_{2n} \\ \vdots \\ \phi_{nn} \end{pmatrix}$$

proportion of variance explained  
(PVE) by Principal Component  $i$

$$\frac{\lambda_i}{\sum_{j=1,n} \lambda_j}$$

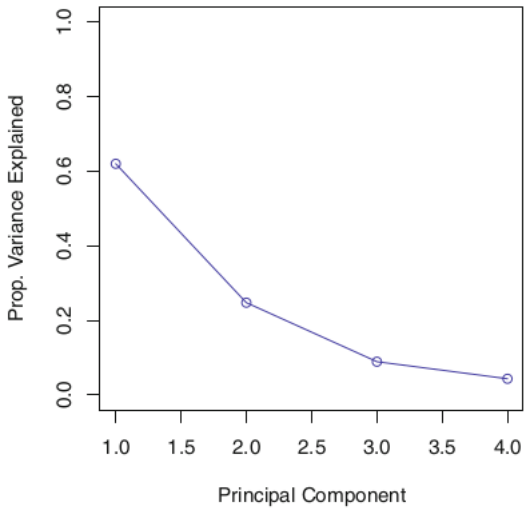
# biplot



example:  
Assault, Murder, Rape  
and Urban Population  
measured on 50 US  
states

|          | PC1       | PC2        |
|----------|-----------|------------|
| Murder   | 0.5358995 | -0.4181809 |
| Assault  | 0.5831836 | -0.1879856 |
| UrbanPop | 0.2781909 | 0.8728062  |
| Rape     | 0.5434321 | 0.1673186  |

## scree plot



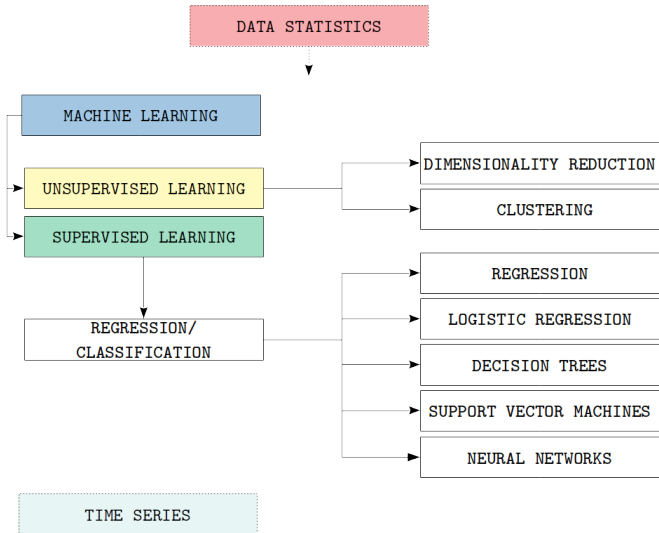
## exercise 2.3

# references

- 'An Introduction to Statistical Learning', Section 6.3 (pages 228-233), 10.2
- 'The Elements of Statistical Learning', Section 14.5.1

# Overview

- 1 Introduction
- 2 Basic statistics
- 3 Multivariate statistics and Principal Component Analysis
- 4 Unsupervised learning: Clustering**
- 5 Supervised learning
- 6 Overview on Time Series





# unsupervised vs supervised learning

- unsupervised learning

$$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \dots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}$$

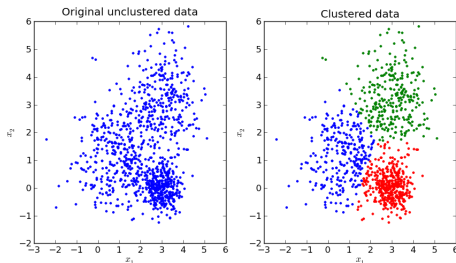
task: to discover interesting things about data **without knowing the true answer**

- supervised learning

$$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \dots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}, Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}$$

task: prediction of the targets

# clustering



- find **clusters** (subgroups) of observations having **similar** features
- clusters  $C_i, i = 1, \dots, K$  are a partition of the observations:

$$C_1 \cup C_2 \cup \dots \cup C_K = \{1, \dots, m\}, \quad C_i \cap C_j = \emptyset, \forall i \neq j$$

- when observations are similar (dissimilar)? domain-specific
- PCA and clustering simplify data via smaller summaries
  - ▶ PCA via low-dimension features representations
  - ▶ clustering via homogenous subgroups among observations

## clustering **methods**

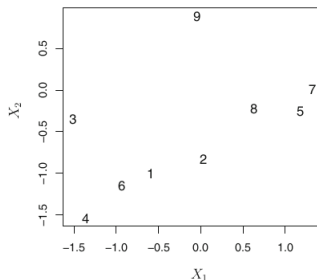
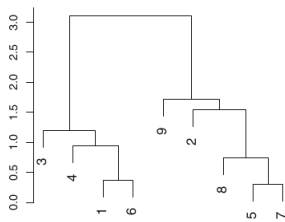
- **hierarchical** clustering:  
the number of clusters is iteratively modified
- **K-means** clustering:  
the number of clusters is decided a priori
- fuzzy clustering:  
observations are assigned to all clusters with a certain probability (not to a single one)

## hierarchical **agglomerative** clustering

- idea: iteratively aggregate  $m$  initial clusters (one per observations) to a final single cluster (including all observations)
- **dendrogram**: attractive tree-based representation of observations and clusters
- the number of clusters is decided ex post
- requires
  - ▶ a point to point distance (**dissimilarity**)  $d(x_i, x_j)$  with  $x_i, x_j \in \mathbb{R}^n$
  - ▶ a cluster to cluster distance (**linkage**)  $D(C_s, C_t)$  with  $s, t \in \{1, \dots, K\}$

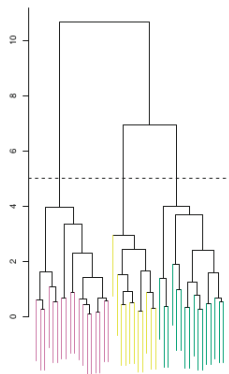
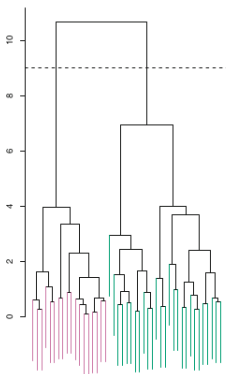
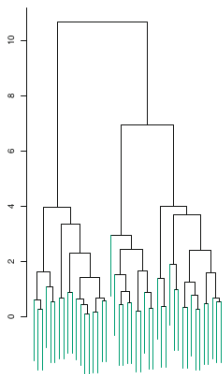
## agglomerative clustering scheme and dendrogram

- 0 select dissimilarity  $d$  and linkage  $D$
- 1 set the  $m$  initial clusters as the  $m$  observations and compute the  $\binom{m}{2} = m(m-1)/2$  pairwise dissimilarities
- 2 for  $i = m, m-1, \dots, 2$ 
  - a identify the pair of closest clusters according to  $D$  (or  $d$  if single observations) and fuse them
  - b compute the pairwise linkage  $D$  for the remaining  $i-1$  clusters



## how to select the final clustering?

- make horizontal cut on the dendrogram
- subgroups below the cut are the clusters
- number of clusters = number of vertical lines crossed by the cut (height of the cut determine the number of clusters  $K$ )



## linkage choices

- SINGLE LINK:  $D(C_s, C_t) = \min_{x_i \in C_s, x_j \in C_t} d(x_i, x_j)$
- COMPLETE LINK:  $D(C_s, C_t) = \max_{x_i \in C_s, x_j \in C_t} d(x_i, x_j)$
- AVERAGE LINK:  $D(C_s, C_t) = \frac{1}{|C_s||C_t|} \sum_{x_i \in C_s, x_j \in C_t} d(x_i, x_j)$
- CENTROID LINK:  $D(C_s, C_t) = d(\bar{x}_s, \bar{x}_t), \quad \bar{x}_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$

example: apply the agglomerative clustering with single link to the 5 points characterized by this pairwise Euclidean distances matrix.

|   | 1  | 2  | 3 | 4 | 5 |
|---|----|----|---|---|---|
| 1 | 0  |    |   |   |   |
| 2 | 9  | 0  |   |   |   |
| 3 | 3  | 7  | 0 |   |   |
| 4 | 6  | 5  | 9 | 0 |   |
| 5 | 11 | 10 | 2 | 8 | 0 |

|       | 1 | 2 | {3,5} | 4 |
|-------|---|---|-------|---|
| 1     | 0 |   |       |   |
| 2     | 9 | 0 |       |   |
| {3,5} | 3 | 7 | 0     |   |
| 4     | 6 | 5 | 8     | 0 |

|         | {1,3,5} | 2 | 4 |
|---------|---------|---|---|
| {1,3,5} | 0       |   |   |
| 2       | 7       | 0 |   |
| 4       | 6       | 5 | 0 |

|         | {1,3,5} | {2,4} |
|---------|---------|-------|
| {1,3,5} | 0       |       |
| {2,4}   | 6       | 0     |



## some considerations on agglomerative clustering

good aspects:

- provides a nested sequence of clusterings (more informative)

bad aspects:

- sensitive to anomalous points/outliers
- irreversible: "bad" early fusions affect the whole structure of nested clusters
- arbitrary fusion choices of equally distant clusters affect the rest of the sequence
- time consuming for large datasets

## K-means clustering

- the number of clusters  $K$  is given
- idea: a "good" clustering is one for which the **within-cluster variation** is as small as possible
- each cluster  $C_k$  is characterized by its **centroid**  $\bar{x}_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$
- tends to minimize the overall distance between each point and its cluster's centroid:

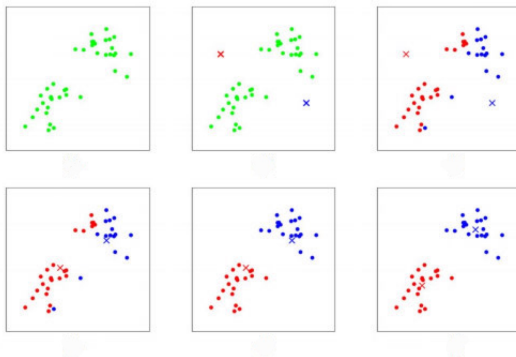
$$\min_{\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K} W(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K) = \sum_{k=1, \dots, K} \sum_{x_j \in C_k} d(x_j, \bar{x}_k)$$

- the algorithm is an alternation of two main steps
  - a assign the points to the cluster with the closest centroid
  - b determine the centroids of each cluster
- the algorithm stops when the two steps do not change the assignment (or do not improve  $W(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K)$ ) anymore



## K-means clustering scheme

- 0 select a distance  $d$  and a number  $K$
- 1 (initialization) randomly determine  $K$  centroids  $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K \in \mathbb{R}^n$
- 2 iterate until the cluster assignment stop changing
  - a (cluster assignments) assign each observation to the closest centroid
  - b (centroids update) update the centroids of each cluster as  $\bar{x}_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$



## some considerations on K-means

- sensitive to anomalous points/outliers
- points can move from a cluster to another
- converges to a sub-optimal solution  $\Rightarrow$  the final solution depends on the random initialization
- if points have not a strong "clustered" structure or  $K$  is "wrong" the algorithm is more sensitive to initialization
- good practice: multiple runs of K-means for different initialization and different  $K$  and select the best solution (in terms of  $W(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_K)$ )

how to verify the quality of a clustering

have we obtained relevant structures? is  $K$  right?

we can perform computational assessments

- by **perturbating** data and verifying if the clustering is stable
  - ▶ add noise to data
  - ▶ delete points (resample without replacement)
  - ▶ bootstrapping (resample with replacement)
- use the **silhouette** measure and plot...

## the silhouette of clusters

suppose point  $i$  belongs to  $C_i$

- mean distance from  $x_i$  to points of cluster  $C$

$$d_{iC} = \frac{1}{|C|} \sum_{x_j \in C} d(x_i, x_j)$$

- mean distance from  $x_i$  to points of its cluster

$$a_i = d_{iC_i}$$

- minimum mean distance from  $x_i$  to points of another cluster

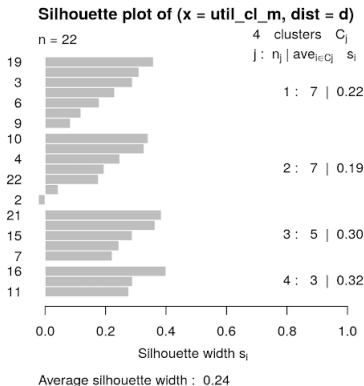
$$b_i = \min_{C \neq C_i} d_{iC}$$

- silhouette of point  $i$

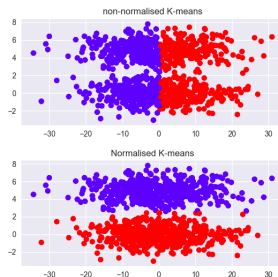
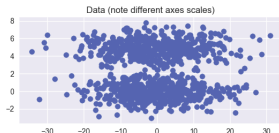
$$SIL_i = \frac{b_i - a_i}{\max\{a_i, b_i\}}$$

- overall silhouette

$$SIL = \frac{1}{m} \sum_{i=1, m} SIL_i$$



## the issue of scaling data



features with different locations or scales may influence the distance of two points (e.g. age and salary) and bias clustering results (the same holds for PCA!)

often it is better to **center** and/or **scale** features

- Z-score scaler:  $\frac{x - \bar{x}}{\sigma} \Rightarrow \text{mean}=0, \text{st.dev.}=1$
- unit scaler:  $\frac{x - \min_x}{\max_x - \min_x} \Rightarrow \text{values in the interval } [0, 1]$

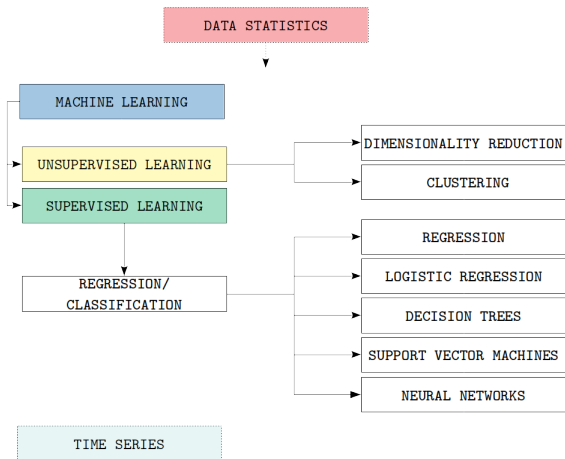
### exercise 3.1

- 'An Introduction to Statistical Learning', Section 10.3



# Overview

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supervised learning  $\equiv$  learning with a **teacher**

idea: observe many features-target pairs to reconstruct the function mapping features to target

# supervised learning

- consider a process producing target  $y$  in correspondence to features  $x \in \mathbb{R}^n$  according to an **unknown** function

$$y = f(x)$$

- supervised learning** determines a model  $m(\cdot)$  approximating  $f(\cdot)$  on the basis of a set of known features-target pairs denoted as **training set** ( $TR$ )

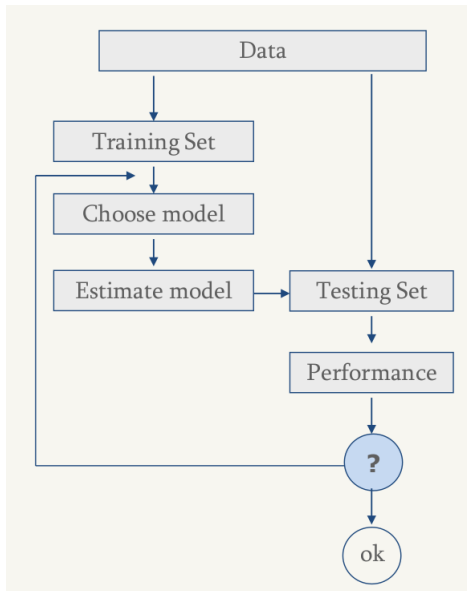
$$TR = \{(x^i, y^i) : x^i \in \mathbb{R}^n, y^i \in ?, i = 1, \dots, m\}$$

- in the **training phase** the parameters of  $m(\cdot)$  are tuned so that the model fits "as much as possible"  $TR$
- once trained the model is tested on a **testing set** ( $TS$ ) of known features-target pairs **not used** in the training, and if "sufficiently good" it can be used to predict the target  $\tilde{y}$  for **unseen** observations  $\tilde{x}$  according to

$$\tilde{y} = m(\tilde{x})$$



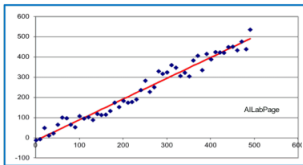
## supervised learning scheme



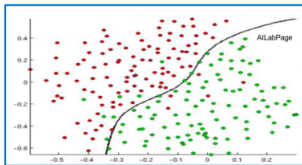
- **generalization**: capability of the trained model  $m(\cdot)$  to correctly predict the target of unseen observations
- **overfitting**: when  $m(\cdot)$  fits "too much" the training data and it is not able to make good predictions for unseen observations

- the form of  $f(\cdot)$  may be of any type
  - ▶ mathematical function
  - ▶ logic formula
  - ▶ the result of an algorithm
  - ▶ a black-box system
  - ▶ ...
- the form of  $m(\cdot)$  may be of different types
  - ▶ mathematical function
  - ▶ a structured object (e.g., a tree, a neural network,...)
  - ▶ a set of rules
  - ▶ ...
- **learning tasks** categorized according to the domain of target  $y$ 
  - ▶  $y$  continuous  $\Rightarrow$  **regression**
  - ▶  $y$  discrete  $\Rightarrow$  **classification** (binary or multi-class)

regression



binary classification



many issues in supervised learning...

- how to select the family of functions of model  $m(\cdot)$
- how to determine **hyperparameters** of  $m(\cdot)$  (model parameters which cannot be tuned during the training phase and must be set a priori)
- which algorithm to adopt in the training phase
- how to split training and testing sets
- how to measure the performance of the trained model
- how to choose/control the model complexity (complex models fit training data well, simple models tend to better generalize)

... many alternative machine learning approaches

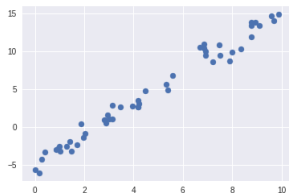
# linear regression

linear regression fast and interpretable

simple linear regression model (single feature)

$$y = \beta_0 + \beta_1 x \text{ (regression line)}$$

- $y \in \mathbb{R}$  (regression task)
- $\beta_0$  intercept
- $\beta_1$  slope



training a simple linear regression consists in determining  $\hat{\beta}_0$  and  $\hat{\beta}_1$  such that the regression line fits better the points

## training simple linear regression

- given a  $TR$  of pairs  $(x^i, y^i)$  with  $i = 1, \dots, m$ , find estimates  $\hat{\beta}_0, \hat{\beta}_1$  such that

$$\hat{\beta}_0 + \hat{\beta}_1 x^i \approx y^i \quad \text{with } i = 1, \dots, m$$

- given  $\hat{\beta}_0, \hat{\beta}_1$ , the prediction  $\hat{y}^i$  for point  $x^i$  is

$$\hat{y}^i = \hat{\beta}_0 + \hat{\beta}_1 x^i$$

and the  $i$ -th error (**residual**) is computed as

$$e^i = y^i - \hat{y}^i$$

- $\hat{\beta}_0, \hat{\beta}_1$  are obtained by solving the following optimization problem where a **loss** function called **residual sum of squares (RSS)** is minimized

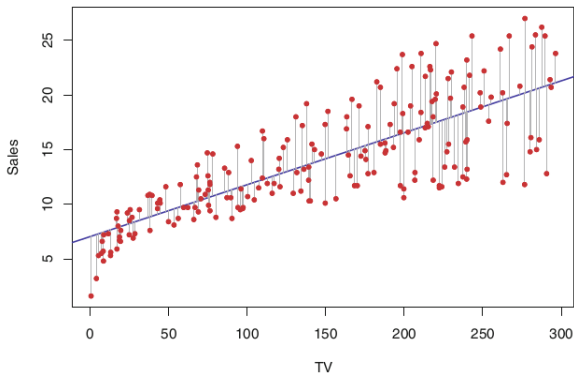
$$\min_{\beta_0, \beta_1 \in \mathbb{R}} \text{RSS} = \sum_{i=1, m} (y^i - \beta_0 - \beta_1 x^i)^2 = \sum_{i=1, m} (y^i - \hat{y}^i)^2 = \sum_{i=1, m} (e^i)^2$$

- $\text{RSS}$  is convex  $\Rightarrow$  the solution is unique: 
$$\begin{cases} \hat{\beta}_1 = \frac{\sum_{i=1, m} (x^i - \bar{x})(y^i - \bar{y})}{\sum_{i=1, m} (x^i - \bar{x})^2} \\ \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \end{cases}$$
- prediction for an unseen  $\tilde{x}$ :  $\tilde{y} = \hat{\beta}_0 + \hat{\beta}_1 \tilde{x}$





example: Sales of a product as a function of the TV advertisings over 200 different markets



$$\hat{\beta}_0 = 7.03, \hat{\beta}_1 = 0.0475 \Rightarrow \text{Sales} = 7.03 + 0.0475 \text{ TV}$$

by spending 1000 more euro in TV advertisings we sell 47.5 more units of product!

**multivariate** linear regression: predictors (features) are  $n$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

- given  $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_n$ , the prediction  $\hat{y}^i$  for point  $x^i \in \mathbb{R}^n$  is

$$\hat{y}^i = \hat{\beta}_0 + \hat{\beta}_1 x_1^i + \hat{\beta}_2 x_2^i + \cdots + \hat{\beta}_n x_n^i$$

- $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_n$  are obtained by solving

$$\begin{aligned} \min_{\beta_0, \beta_1, \beta_2, \dots, \beta_n} \text{RSS} &= \sum_{i=1, m} (y^i - \beta_0 - \beta_1 x_1^i - \beta_2 x_2^i - \cdots - \beta_n x_n^i)^2 = \\ &= \sum_{i=1, m} (y^i - \hat{y}^i)^2 = \sum_{i=1, m} (e^i)^2 \end{aligned}$$

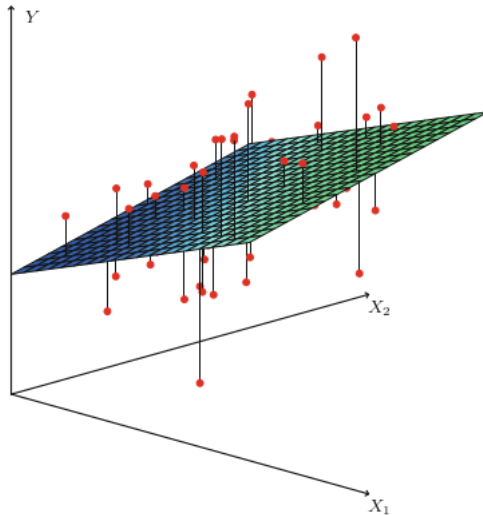
- the solution  $\hat{\beta} \in \mathbb{R}^n$  can be computed as

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

- prediction for an unseen  $\tilde{x}$ :  $\tilde{y} = \hat{\beta}_0 + \hat{\beta}_1 \tilde{x}_1 + \hat{\beta}_2 \tilde{x}_2 + \cdots + \hat{\beta}_n \tilde{x}_n$



## a 2-dimensional linear regression



how to measure the regression accuracy (on the training or testing set)?

- residual sum of squares ( $RSS$ )

$$RSS = \sum_{i=1,m} (y^i - \hat{y}^i)^2$$

- mean square error ( $MSE$ )

$$MSE = \frac{1}{m} \sum_{i=1,m} (y^i - \hat{y}^i)^2$$

(root mean square error ( $RMSE$ ) =  $\sqrt{MSE}$ )

- coefficient of determination ( $R^2$ )

$$\text{total sum of squares } TSS = \sum_{i=1,m} (y^i - \bar{y})^2$$

$$R^2 = 1 - \frac{RSS}{TSS}$$

# accuracy of the regression coefficients

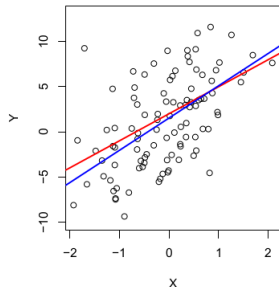
simple linear regression coefficients  $\hat{\beta}_0, \hat{\beta}_1$  are determined on *TR* data **actually** generated by

$$y = \beta_0 + \beta_1 x + \epsilon$$

where  $\epsilon$  is an error term considering:

- the probable not linear nature of the relationship between  $y$  and  $x$
- there may be other variables influencing  $y$
- there may be measurement errors
- ...

$\Rightarrow$  regression coefficients  $\hat{\beta}_0, \hat{\beta}_1$  differ from **real** coefficients  $\beta_0, \beta_1$



example:

**red line** is given by

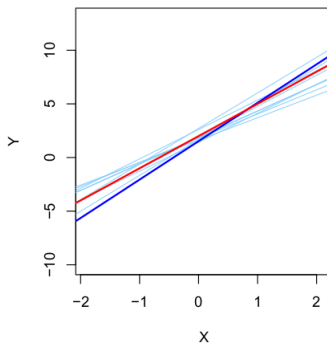
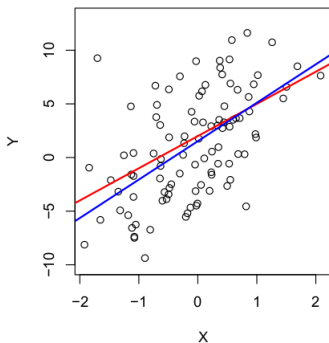
$$y = 2 + 3x$$

**blue line** is the regression line of observed data generated by a process of the type

$$y = 2 + 3x + \epsilon$$

regression line is an **unbiased estimator**:

$\beta_0, \beta_1$  **tend** to be not overestimated/underestimated by  $\hat{\beta}_0, \hat{\beta}_1$



how well regression coefficients  $\hat{\beta}_0, \hat{\beta}_1$  approximate unknown real  $\beta_0, \beta_1$ ?

the average of  $\hat{\beta}_0, \hat{\beta}_1$  determined over many datasets would be accurate but we have only one  $TR...$

assuming all  $\epsilon_i$  (associated to  $x^i$ ) having common standard deviation  $\sigma_\epsilon$  and uncorrelated, we can estimate the variance of  $\hat{\beta}_0, \hat{\beta}_1$ :

it can be shown that

$$\sigma_{\hat{\beta}_0}^2 = \sigma_\epsilon^2 \left[ \frac{1}{m} + \frac{\bar{x}^2}{\sum_{i=1}^m (x^i - \bar{x})^2} \right] \quad \text{and} \quad \sigma_{\hat{\beta}_1}^2 = \frac{\sigma_\epsilon^2}{\sum_{i=1}^m (x^i - \bar{x})^2}$$

- $\sigma_{\hat{\beta}_1}^2$  tends to be smaller for larger spread of  $x^i$
- $\sigma_{\hat{\beta}_0}^2$  would be the same as the one for the sample mean if  $\bar{x} = 0$
- if  $\sigma_\epsilon$  is unknown it can be estimated as  $\sqrt{RSS/(m-2)}$  (residual standard error)
- $\sigma_{\hat{\beta}_0}^2, \sigma_{\hat{\beta}_1}^2$  can be used to construct confidence intervals, for example:  
95% confidence intervals for  $\hat{\beta}_0$  and  $\hat{\beta}_1$  are

$$\left[ \hat{\beta}_0 - 2\sigma_{\hat{\beta}_0}, \hat{\beta}_0 + 2\sigma_{\hat{\beta}_0} \right] \quad \text{and} \quad \left[ \hat{\beta}_1 - 2\sigma_{\hat{\beta}_1}, \hat{\beta}_1 + 2\sigma_{\hat{\beta}_1} \right]$$

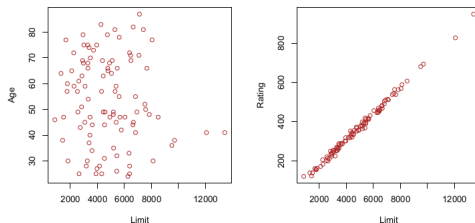


# collinearity and multicollinearity

uniqueness of regression coefficients is important for interpretation...

**collinearity** refers to the situation in which two or more predictor variables are closely related to one another

example: given a dataset of credit card holders, determine variable balance (credit card debt for each individual) based on predictors age, limit (credit limit), and rating (credit rating)



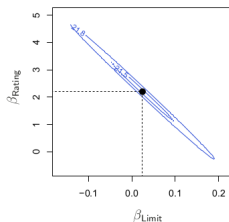
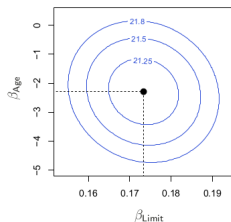
- age and limit exhibit no correlation
- rating and limit are highly correlated  $\implies$  they are **collinear**!



since collinear predictors vary together it is difficult to determine their real single influence on the target variable...

consider the balance regression on age and limit (Model 1), and the balance regression on rating and limit (Model 2)

RSS **level curves** plotted for the range  $\left[\hat{\beta} - 4\sigma_{\beta}, \hat{\beta} + 4\sigma_{\beta}\right]$  for every predictor:



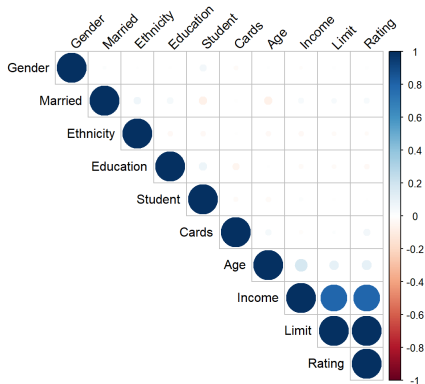
|         |           | Coefficient | Std. error |
|---------|-----------|-------------|------------|
| Model 1 | Intercept | -173.411    | 43.828     |
|         | age       | -2.292      | 0.672      |
|         | limit     | 0.173       | 0.005      |
| Model 2 | Intercept | -377.537    | 45.254     |
|         | rating    | 2.202       | 0.952      |
|         | limit     | 0.025       | 0.064      |

- the collinear rating-limit has level curves characterized by "narrow valleys"
- the range of limit in the collinear case is much larger
- small data change can move the regression coefficients anywhere along the narrow valley

⇒ much more uncertainty of the regression coefficients in the collinear case!



collinearity can be detected by the **correlation matrix** inspection



if two predictors are highly correlated (above a certain threshold) one can be removed from the regression model without compromising accuracy of the regression fit

**multicollinearity** is the collinearity between three or more predictors even if no pair of predictors has a particularly high correlation

⇒ multicollinearity cannot be captured by the correlation matrix

multicollinearity is detected by the **variance inflation factor** (VIF):

$$\text{VIF}(\hat{\beta}_j) = \frac{1}{1 - R^2_{x_j|x_{-j}}}$$

where  $R^2_{x_j|x_{-j}}$  is the  $R^2$  obtained by the regression of predictor  $x_j$  on all other predictors ( $x_{-j}$ )

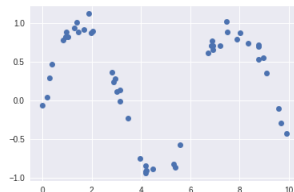
- VIF is always larger (or equal) than 1
- in practice VIF exceeding 5 or 10 indicate a collinearity problem

in the balance regression  $\text{VIF}(\text{age})=1.01$ ,  $\text{VIF}(\text{rating})=160.67$ , and  $\text{VIF}(\text{limit})=160.59$

# polynomial regression

sometimes linear regression is not suited...

...but linearity is desirable (simple training problem)

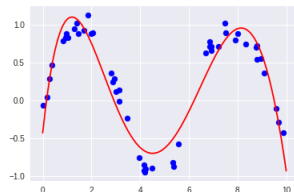


- idea: maintain the linearity of the model with respect to the training variables  $\beta$  and polynomially map the features in a larger space (of dimension  $q$ )
- example of  $q = 3$  **polynomial transformation**  $x \Rightarrow \begin{pmatrix} x \\ x^2 \\ x^3 \end{pmatrix}$

$$y = \beta_0 + \beta_1 x \Rightarrow y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$$

features are nonlinearly transformed (in a 3D space) but the model remains linear in parameters  $\beta$ !

exercise 4.1



# model validation

typical supervised learning scheme

- 1 selection of the class of model  $m(\cdot)$  (e.g. polynomial regression)
- 2 hyperparameters selection (e.g. polynomial degree)
- 3 training the model on  $TR$
- 4 use of the trained model to make prediction on unseen new data

steps (1) and (2) crucial for the prediction performance of the model

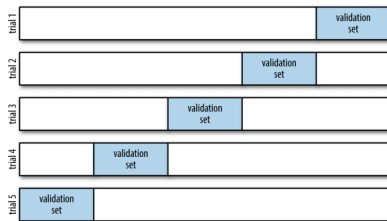
- how to select the model?
- how to select hyperparameters?

a model **validation** is needed!



## model validation

- **holdout set**: remove a subset of available data to create a testing set(not used in the training!)  
if testing set small scarce significance, if big too much data substracted to the training
- **k-fold cross-validation**: train the model  $k$  times (trials or folds), at each trial the dataset is divided in a training and a **validation set** (used to measure the accuracy), the validation set is a  $\frac{1}{k}$  fraction of the number of available data, the final performance is the average over all  $k$  folds



- ▶ all data considered one time in the validation
- ▶ time consuming

which model/hyperparameters to select?

optimization of the model/hyperparameters is too expensive!

model/hyperparameters selection done heuristically via **grid-search**

- 1 for every hyperparameter (including model structure) choose a set of reasonable values (all possibly)
- 2 perform a k-fold cross-validation for every combination of hyperparameters values
- 3 select the hyperparameters values in which the best model accuracy is achieved

example: polynomial regression  $\Rightarrow \begin{cases} q : & \{1, 2, 3\} \\ \text{fit\_intercept} : & \{True, False\} \end{cases}$

|           |           |           |
|-----------|-----------|-----------|
| (1,True)  | (2,True)  | (3,True)  |
| (1,False) | (2,False) | (3,False) |

very time consuming!

`sklearn.model_selection.GridSearchCV()`

# improve model performance

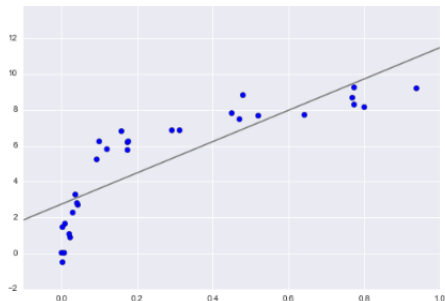
what if the trained model does not work well?

- adopt a more complex/expressive model?
- adopt a less complex/expressive model?
- collect more training data?
- increase the number of features?
- decrease the number of features?
- improve the dataset quality?

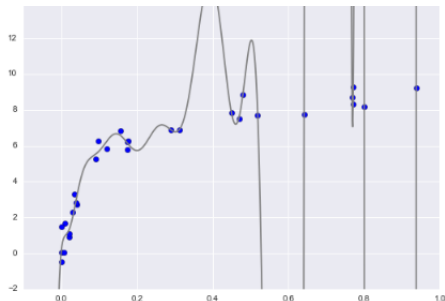
often in machine learning the answer is counterintuitive!



let's focus on model complexity

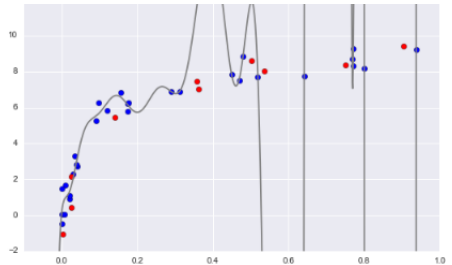
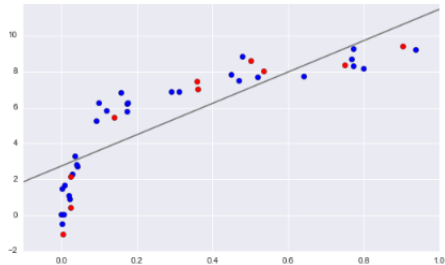


linear model does not fit data as data complexity is higher than the model complexity, **underfitting** occurs  $\Rightarrow$  **bias** model



highly nonlinear model has enough complexity to fit data but it does not describe well the process, **overfitting** occurs  $\Rightarrow$  **variance** model

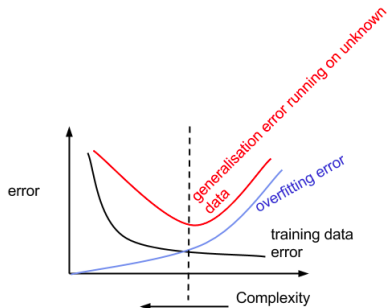
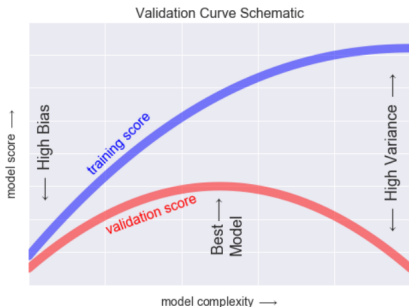
let's focus on model complexity



neither of the two models is able to predict the testing points!

- for highly bias models low accuracy on both training and testing sets
- for highly variance models high accuracy on training set, low on testing set

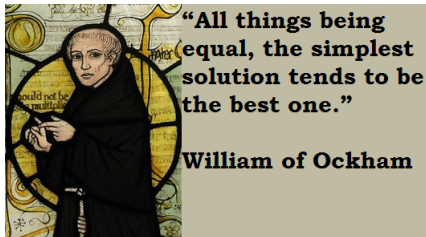
a **trade-off** between bias and variance is required!



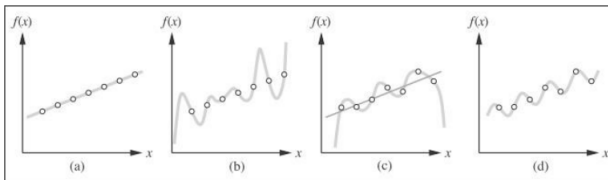
the goal is to obtain a model near the maximum of the validation score curve!

grid-search+cross-validation may help...

follow the Ockham's razor principle...



prefer the **simplest** hypothesis **consistent** with data!



data are often in a format not suited for the learning...

data **preprocessing** may help in improving the prediction model

- image transformation (e.g. pixel matrix for handwritten digit)
- categorical data transformations
  - ▶ bad method: transform categories into integer numbers  
L'Aquila  $\rightarrow$  1, Rome  $\rightarrow$  2, Milan  $\rightarrow$  3  
inherits a meaningless order (L'Aquila  $<$  Rome  $<$  Milan)
  - ▶ good method: **one-hot encoding** transforms categorical data in array with a number of binary elements equal to the number of categories (-1)  
L'Aquila  $\rightarrow$  [1 0], Rome  $\rightarrow$  [0 1], Milan  $\rightarrow$  [0 0]  
features grow too much with many categories
- text data transformations
  - ▶ **words count**: count the number of occurrences of each word in a text sample = ['problem of evil', 'evil queen', 'horizon problem']

| evil | horizon | of | problem | queen |
|------|---------|----|---------|-------|
| 1    | 0       | 1  | 1       | 0     |
| 1    | 0       | 0  | 0       | 1     |
| 0    | 1       | 0  | 1       | 0     |

- derivations (e.g. polynomial transformation)
- NA values imputations
- standardization/normalization (**target normalization** very important in regression!)

`sklearn.preprocessing`, `sklearn.feature_extraction`



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**feature selection** may be useful to limit the complexity by reducing the actually used features

all possible subsets of features are  $2^n$ , too expensive training  $2^n$  models...

...heuristics!

- **forward selection**

- 1 start with the null-model, train all  $n$  single feature models and add to the null-model the feature corresponding to the best performing single feature model
- 2 train all the  $n - 1$  two features models and add to the previous single features model the feature corresponding to the best performing two features model
- 3 iterate until some stopping condition is reached

- **backward selection**

- 1 start with whole  $n$  feature model and train it
- 2 iteratively remove the feature which is "statistically less significant" until some stopping condition is reached

- **mixed selection**: combine forward and backward selection

`sklearn.feature_selection`



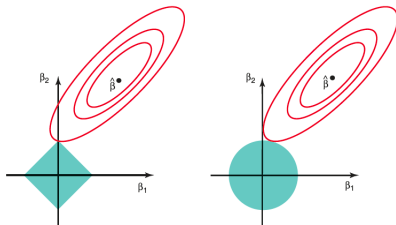
complexity can be also controlled by adding a **regularization** term to the loss function

example: multivariate regression

$$\min \quad \text{loss} + \lambda \cdot \text{regularization} = \sum_{i=1,m} (y^i - \hat{y}^i)^2 + \lambda \|\beta\|_p^p$$

typical choices

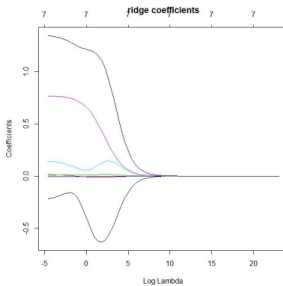
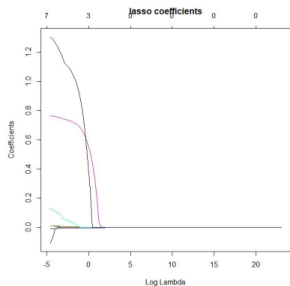
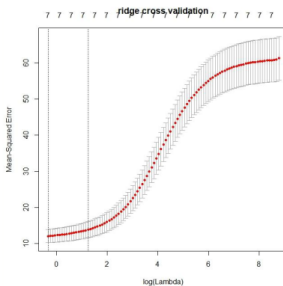
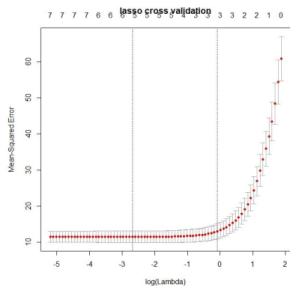
- $p = 1$ : **lasso** regression  $\rightarrow \min \sum_{i=1,m} (y^i - \hat{y}^i)^2 + \lambda \sum_{j=0,n} |\beta_j|$
- $p = 2$ : **ridge** regression  $\rightarrow \min \sum_{i=1,m} (y^i - \hat{y}^i)^2 + \lambda \sum_{j=0,n} |\beta_j|^2$



lasso  $\Rightarrow$  more **sparse** models

ridge  $\Rightarrow$  more **smooth** models

# example: grid-search cross-validation results for lasso regression and ridge regression





check the dataset quality:

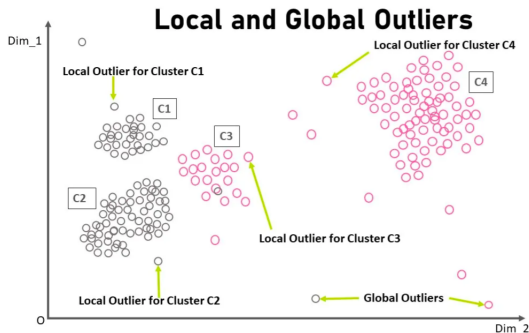
- are there missing values? How to remove them?
  - ▶ remove rows with missing values
  - ▶ replace missing values with representative values (mean, mode, median,...)
  - ▶ replace missing values with "close" values
- are data informative?
  - ▶ collinearity and multicollinearity
- are there anomalous values?
  - ▶ outlier-novelty detection

# outliers detection

let's focus on anomalies...

anomalies are "polluted" samples in the training set (**outliers**) or in unseen observations (**novelties**)

- **global outliers**: samples significantly different from the rest of the dataset
- **local outliers**: samples significantly different from their neighbours

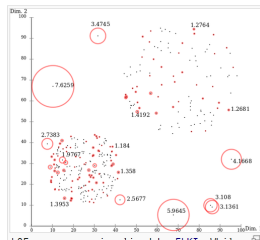
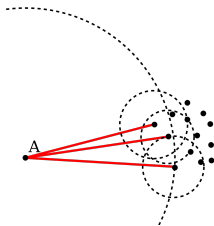


## Local Outlier Factor (LOF)

- consider set  $A(x^i; K)$  of observations of the **K nearest neighbors** from a sample  $x^i$
- compute the **density** of  $x^i$  as its maximum distance from the farthest observation in  $A(x^i; K)$ :

$$\text{den}(x^i; K) = \max_{x^j \in A(x^i; K)} d(x^i, x^j)$$

- if ratio  $\text{LOF}(x^i; K) = \frac{\text{den}(x^i; K)}{\frac{1}{|A(x^i; K)|} \sum_{x^j \in A(x^i; K)} \text{den}(x^j; K)} \geq \eta \gg 1 \implies x^i$  is an outlier

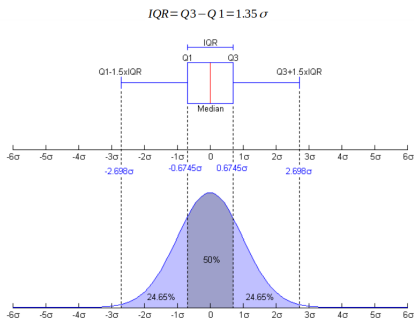


## LOF considerations

- it is able to determine local outliers and not only global
- performance affected by  $K$
- performance affected by  $\eta$

by assuming normally distributed data  **$\sigma$ -clipping** can be used to detect outliers

- for normal distributions  $IQR = 1.35\sigma$
- $\sigma$ -clipping idea: exploit **observed** Q1 and Q3 to estimate  $\sigma$  and construct confidence interval for not outliers (**inliers**)



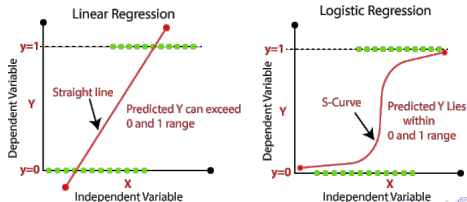
## exercise 4.2

# logistic regression

intermediate between regression and binary classification tasks

- consider a binary classification dataset ( $y^i \in \{0, 1\}$ ) with single feature, assume we want to estimate for each feature  $x^i$  the probability  $p^i = P(y^i = 1)$
- in principle one could fit a line to data (linear regression) and set  $p(x^i) = \hat{\beta}_0 + \hat{\beta}_1 x^i$  but  $p(x^i)$  is not in the interval  $[0, 1]$ ...
- better to fit a S-shaped nonlinear function ranging in the interval  $[0, 1]$  like

$$p(x) = \frac{1}{1 + e^{-\beta_0 - \beta_1 x}}$$



- logistic regression training consists in determining the parameters estimates  $\hat{\beta}_0, \hat{\beta}_1$  by maximizing the **likelihood function** over  $TR$

$$\ell(\beta_0, \beta_1) = \prod_{i: y^i=1} p(x^i) \prod_{i: y^i=0} (1 - p(x^i))$$

- once trained the model one can use a threshold  $\rho$  (typically  $\rho = 0.5$ ) to classify data according to

$$\tilde{y} = \begin{cases} 1 & \text{if } p(\tilde{x}) \geq \rho \\ 0 & \text{if } p(\tilde{x}) < \rho \end{cases}$$

- logistic regression easily extends to the multivariate case

$$p(x) = \frac{1}{1 + e^{-\beta_0 - \beta_1 x_1 - \beta_2 x_2 - \dots - \beta_n x_n}}$$

how to measure binary classification performance?

assume targets are  $\{\text{positive}, \text{negative}\}$  and the number of samples of the two classes are  $P$  and  $N$

confusion matrix

|                 | predicted positive  | predicted negative  |
|-----------------|---------------------|---------------------|
| actual positive | true positive (TP)  | false negative (FN) |
| actual negative | false positive (FP) | true negative (TN)  |

accuracy score (ACC):  $\frac{TP+TN}{P+N}$

ACC not suited for imbalanced datasets...

"a dataset is imbalanced when there is a significant disproportion among the number of samples of each class of the problem"

"trivial" example: assuming 99% of samples is  $\ominus$  and only 1% is  $\oplus$ , a model classifying every sample as  $\ominus$  achieve a 99% accuracy but is useless!



data imbalance is widespread (e.g. medical applications, spam detection,...)  
alternative performance metrics can be considered to deal with imbalance  
assume we are more interested in the minority class  $P$ ...

confusion matrix

|                 | predicted positive  | predicted negative  |
|-----------------|---------------------|---------------------|
| actual positive | true positive (TP)  | false negative (FN) |
| actual negative | false positive (FP) | true negative (TN)  |

- $TPR = \frac{TP}{P}$  (true positive rate, **sensitivity**, **recall**)
- $TNR = \frac{TN}{N}$  (true negative rate, **specificity**)
- $FPR = \frac{FP}{N}$  (false positive rate)
- $FNR = \frac{FN}{P}$  (false negative rate)
- $PPV = \frac{TP}{TP+FP}$  (positive predicted value, **precision**)
- $NPV = \frac{TN}{TN+FN}$  (negative predicted value)
- $BA = \frac{TPR+TNR}{2}$  (**balanced accuracy**)



let's focus on **precision**...

$$PPV = \frac{\# \text{ correct predicted P}}{\# \text{ total predicted P}} = \frac{TP}{TP+FP}$$

- precision is zero if all samples are predicted as  $\ominus$
- precision is cheated by predicting as  $\oplus$  only a small number of samples for which I am pretty sure ( $FP \rightarrow 0 \implies PPV \rightarrow 100\%$ )

let's focus on **recall**...

$$TPR = \frac{\# \text{ correctly predicted P}}{\# \text{ total P}} = \frac{TP}{TP+FN}$$

- recall is zero if all samples are predicted as  $\ominus$
- recall is cheated by predicting all samples as  $\oplus$  ( $FN = 0 \implies TPR = 100\%$ )

cheating high precision  $\implies$  bad recall (high  $FN$ )

cheating high recall  $\implies$  bad precision (high  $FP$ )

**F1 score**: precision-recall harmonic mean (correctness-coverage trade-off)

$$F1 = 2 \frac{PPV \cdot TPR}{PPV + TPR} = \frac{TP}{TP + \frac{1}{2}(FP + FN)}$$



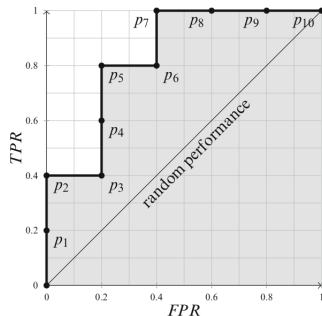
when a threshold parameter is used to transform a classification score into binary classification (e.g. logistic regression), two issues may rise:

- the informative "probabilistic" interpretation of the classification score is ignored
- threshold is arbitrary

**ROC** (receiver operating characteristic) curve allows to evaluate such classifiers without selecting a specific threshold

- rank samples based on the score
- plot FPR and TPR for the threshold ranging from most restrictive to least restrictive

| Rank | Score | Actual class | FPR | TPR | ROC point |
|------|-------|--------------|-----|-----|-----------|
| #1   | 1.0   | Positive     | 0.0 | 0.2 | $p_1$     |
| #2   | 0.9   | Positive     | 0.0 | 0.4 | $p_2$     |
| #3   | 0.85  | Negative     | 0.2 | 0.4 | $p_3$     |
| #4   | 0.7   | Positive     | 0.2 | 0.6 | $p_4$     |
| #5   | 0.6   | Positive     | 0.2 | 0.8 | $p_5$     |
| #6   | 0.45  | Negative     | 0.4 | 0.8 | $p_6$     |
| #7   | 0.35  | Positive     | 0.4 | 1.0 | $p_7$     |
| #8   | 0.3   | Negative     | 0.6 | 1.0 | $p_8$     |
| #9   | 0.2   | Negative     | 0.8 | 1.0 | $p_9$     |
| #10  | 0.05  | Negative     | 1.0 | 1.0 | $p_{10}$  |



**AUC** (area under the curve) score of a classifier is the probability to assign higher classification score to a random  $\oplus$  sample with respect to a random  $\ominus$  sample

exercise 4.3



# dealing with data imbalance

how to deal with data imbalance problems? two main strategies...

- **cost-sensitive classifiers:**

during the training phase different weights are given to samples according to their class membership

- **data level preprocessing:**

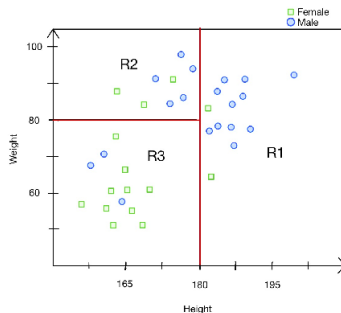
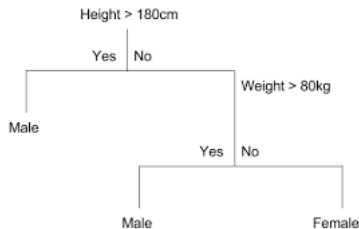
- 1 undersampling: a subset of samples of the majority class is removed
  - ★ may discard potentially relevant samples
- 2 oversampling: a subset of **new** minority class samples is added by replicating samples or by "estimating" new ones (SMOTE method)
  - ★ may cause overfitting
- 3 hybrid: undersampling+oversampling

imbalanced-learn package

# decision trees

machine learning model for **classification** (binary and multiclass) and regression

- easily interpretable models and representable with a tree structure (widely used in medical applications)
- idea: the tree represents a sequence of **splitting** rules operated on the features' domain
- **branch** nodes (splitting decisions) and **leaf** nodes (final target assignment)
- the class of leaf nodes is the one of the largest number of training observations falling into the node



- multiclass classification context

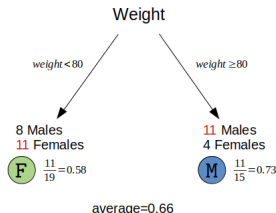
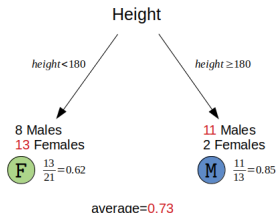
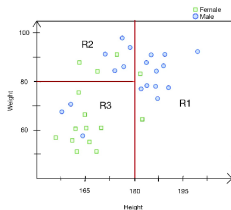
$$TR = \{(x^i, y^i) : x^i \in \mathbb{R}^n, y^i \in \{1, 2, \dots, K\}, i = 1, \dots, m\}$$

- given feature  $x_j$  and **threshold**  $s$ , the splitting **regions**  $R_1(j, s), R_2(j, s)$  are

$$R_1(j, s) = \{x : x_j < s\} \text{ and } R_2(j, s) = \{x : x_j \geq s\}$$

no need of threshold for binary features!

- the class of a region is its majority class
- the "quality" of a splitting is the ability to separate observations of different classes according the **purity** ratio  $\frac{\# \text{ majority class}}{\# \text{ observations in the region}}$

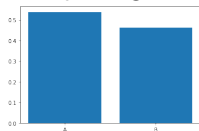


often entropy is used as **impurity** measure (instead of the purity ratio)

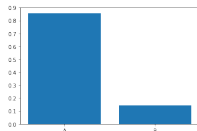
$$H = - \sum_{k=1, K} f_k \log(f_k)$$

where  $f_k$  is the frequency of occurrences of class  $k$  in the region

bad splitting



good splitting



how to train a **classification tree**?

in principle determine regions and thresholds  $R_\ell(j, s)$  with  $j = 1, \dots, n, \ell = 1, \dots, M$  minimizing the overall entropy of all splittings

too expensive to consider every partition of the feature space in  $M$  regions!

⇒ a top-down **greedy** approach: **recursive binary splitting**

- 1 at each step determine feature  $x_j$  and threshold  $s$  such that the splitting

$$R_\ell(j, s) = \{x : x_j < s\} \text{ and } R_{\ell+1}(j, s) = \{x : x_j \geq s\}$$

minimize the average entropy of the two formed regions among all possible features and thresholds

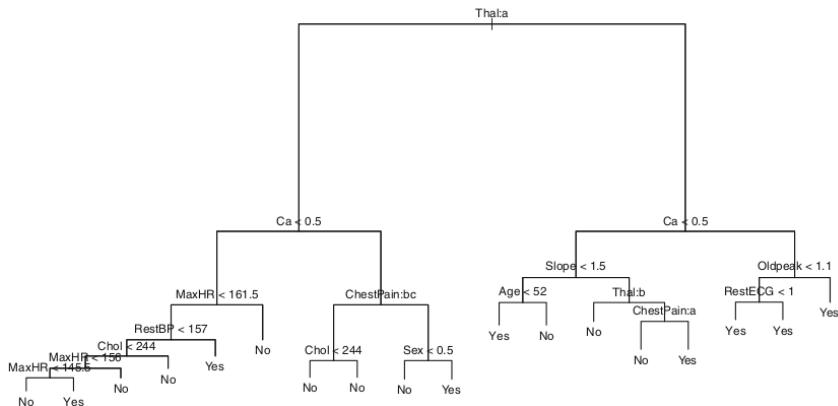
- 2 stop if some condition is reached (e.g. number of points in a region, number of nodes, entropy of the splittings)



## recursive binary splitting

- greedy  $\Rightarrow$  at each step the best current split is added not considering the split leading to the best final tree
- computing the best current split ( $x_j$  and  $s$ ) can be done efficiently
- can be used for mixed features (continuous, categorical,...)

example: dataset of 303 patients with binary target indicating the presence of heart disease (the specified value in the branch node corresponds to left branch)



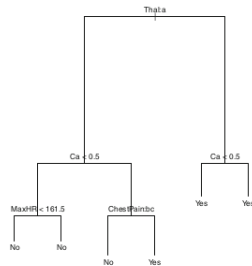
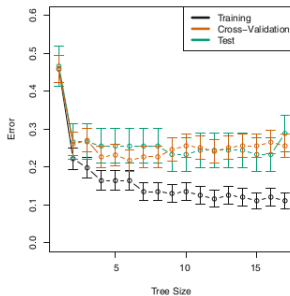
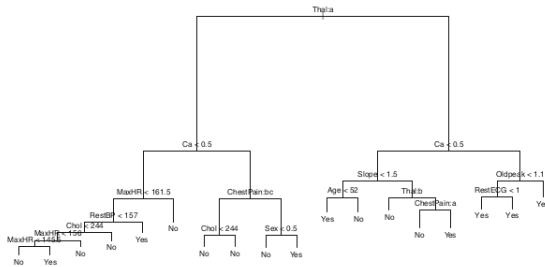
- recursive binary splitting may overfit data with too complex trees
- smaller trees may have lower variance with slightly larger bias
- bad strategy  $\Rightarrow$  stop splits early when the entropy reduction is below an high threshold ('better' splits can occur later in the tree)
- good strategy  $\Rightarrow$  **pruning**:
  - 1 grow a large initial tree  $T_0$  where  $|T_0|$  is the number of leaf nodes
  - 2 for different values of coefficient  $\alpha$  determine a sequence of sub-trees  $T \subseteq T_0$  minimizing

$$\sum_{p=1}^{|T|} \left( - \sum_{k=1, K} f_k \log(f_k) \right) + \alpha |T|$$

- 3 operate a cross-validation to determine the best  $\alpha$  value and so the best sub-tree



# a pruned classification tree



## some decision trees considerations

- Gini coefficient can be used as a impurity measure
- in the regression version the predicted target of leaf nodes is the mean value of training observations falling in that node
- decision trees are close to human decision-making mechanisms
- are easy to depict and interpret
- can deal with mixed features
- no high level of accuracy compared to other models
- not robust to data

⇒ to overcome the previous drawbacks multiple trees are combined together  
(**random forests**)

### exercise 4.4



# Support Vector Machines

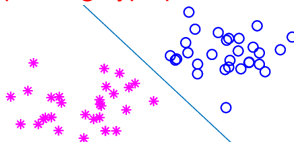
SVMs are machine learning models for binary classification (adaptable to multiclass and regression)

- binary classification problem

$$TR = \{(x^i, y^i) : x^i \in \mathbb{R}^n, y^i \in \{-1, +1\}, i = 1, \dots, m\}$$

- assume  $TR$  is **linearly separable**, i.e. the points of the two classes can be separated by a  $(n - 1)$ -dimensional **separating hyperplane**

$$w^T x + b = 0, \quad w \in \mathbb{R}^n, b \in \mathbb{R}$$



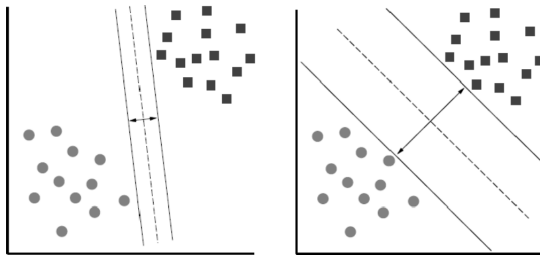
- training a SVM  $\Rightarrow$  determine optimal parameters  $(\hat{w}, \hat{b})$  of the hyperplane separating samples in  $TR$  according to their class
- once trained the model, the **decision function**  $h(\cdot)$  to classify unseen observations is

$$h(\tilde{x}) = \text{sign}(\hat{w}^T \tilde{x} + \hat{b})$$

- if  $TR$  is linearly separable, a separating hyperplane ( $w, b$ ) divides data with a tolerance gap  $\rho$  according to the **separating conditions**

$$\begin{cases} w^T x^i + b \geq \rho & \text{for } y^i = +1 \\ w^T x^j + b \leq -\rho & \text{for } y^j = -1 \end{cases} \Rightarrow \begin{cases} w^T x^i + b \geq 1 & \text{for } y^i = +1 \\ w^T x^j + b \leq -1 & \text{for } y^j = -1 \end{cases}$$

- the separating hyperplane are infinitely many! which one is better?



- the **maximum margin** hyperplane (the one with maximum distance from the closest points) is better!
  - hyperplanes close to training samples are more sensitive to noise
  - intuitively/theoretically a big margin hyperplane is more likely to better classify unseen data

a maximum margin training problem...

- the distance of a point  $x^i$  from hyperplane  $(w, b)$  is

$$d(x^i; w, b) = \frac{|w^T x^i + b|}{\|w\|}$$

- the margin of a hyperplane  $(w, b)$  for  $TR$  is

$$\rho(w, b) = \min_{i=1, \dots, m} \left\{ \frac{|w^T x^i + b|}{\|w\|} \right\}$$

- it can be shown that: maximizing  $\min_{i=1, \dots, m} \left\{ \frac{|w^T x^i + b|}{\|w\|} \right\} \equiv$  minimizing  $\|w\|$
- SVM training optimization problem

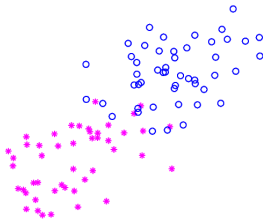
$$\min_{w \in \mathbb{R}^n, b \in \mathbb{R}} \frac{\|w\|^2}{2}$$

$$\text{subject to } y^i(w^T x^i + b) \geq 1 \quad i = 1, \dots, m$$

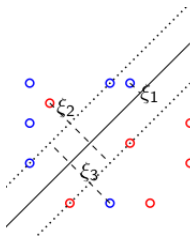
- it is a convex problem!



what if  $TR$  is **not linearly separable**?



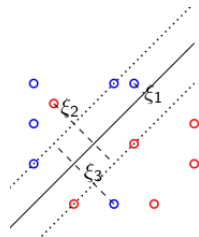
- in such a case there are no separating hyperplanes but we may wish to maintain linearity of the separating surface
- idea: for each separating constraint  $y^i(w^T x^i + b) \geq 1$  add a **slack variable**  $\xi_i$  with  $\xi_i \geq 0$  allowing a **potential** violation of it



## the SVM soft-margin formulation

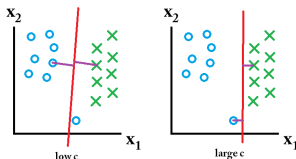
$$\min_{w \in \mathbb{R}^n, b \in \mathbb{R}, \xi \in \mathbb{R}^m} \frac{\|w\|^2}{2} + C \sum_{i=1}^m \xi_i$$

$$\text{subject to } y^i(w^T x^i + b) \geq 1 - \xi_i \quad i = 1, \dots, m$$
$$\xi_i \geq 0 \quad i = 1, \dots, m.$$

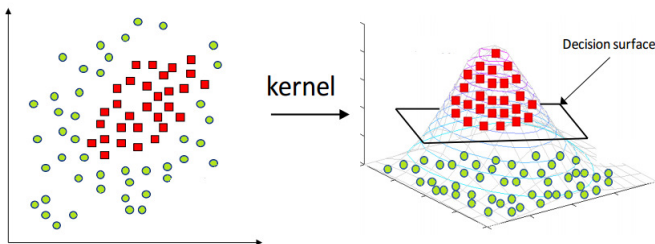


- $\xi_i$  measures how much constraint  $i$  is violated
- $0 \leq \xi_i < 1$  the point is correctly classified although violating the constraint
- $\xi_i \geq 1$  the point is misclassified
- $C$  is the weight of the term penalizing the misclassifications in the objective
- it is still a convex problem!

$C$  values comparison



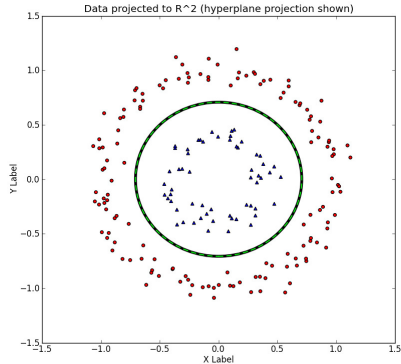
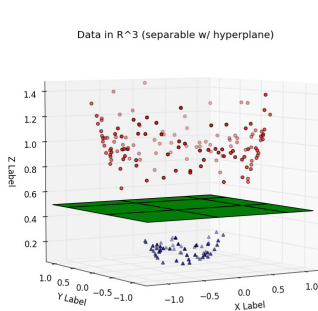
what if  $TR$  is not suited for a linear decision function?



- idea: mapping input data  $x$  from their original **input space** to a higher dimensional space (**feature space**) where data are more likely to be linearly separable
- the nonlinear mapping  $\phi : \mathbb{R}^n \rightarrow \mathcal{H}$  has 2 main peculiarities
  - ▶  $\dim(\mathcal{H}) \gg n$
  - ▶ it is known an analytical formula to efficiently compute  $\phi(x^i)^T \phi(x^j)$  (**kernel**)



a linear separating surface in the feature space is nonlinear in the original input space



example:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \xrightarrow{\phi} \begin{pmatrix} -x_1 \\ x_2 \\ -x_1^2 \end{pmatrix} \equiv \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}$$

$$-w_1x_1 + w_2x_2 - w_3x_1^2 + b = 0 \quad \leftarrow \quad w_1z_1 + w_2z_2 + w_3z_3 + b = 0$$

the **nonlinear** SVM formulation

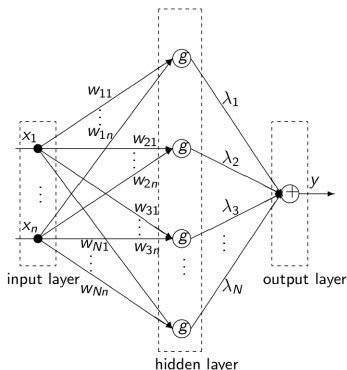
$$\begin{aligned} \min_{w \in \mathbb{R}^{dim(\mathcal{H})}, b \in \mathbb{R}, \xi \in \mathbb{R}^m} \quad & \frac{\|w\|^2}{2} + C \sum_{i=1}^m \xi_i \\ \text{subject to} \quad & y^i (w^T \phi(x^i) + b) \geq 1 - \xi_i \quad i = 1, \dots, m \\ & \xi_i \geq 0 \quad i = 1, \dots, m \end{aligned}$$

- it is still a convex problem!
- the explicit mapping  $\phi$  may not be known or computable (the feature space may be of infinite dimension)  $\Rightarrow$  in the **dual** reformulation of the problem mapping  $\phi$  appears only as kernel terms  $\phi(x^i)^T \phi(x^j)$  which can be efficiently computed (**kernel trick**)
- typical kernel functions:
  - ▶ linear:  $x^i{}^T x^j$  (corresponding to a linear SVM)
  - ▶ gaussian (rbf):  $e^{-\gamma \|x^i - x^j\|^2}$  for  $\gamma > 0$
  - ▶ polynomial:  $(x^i{}^T x^j + 1)^p$  for  $p \geq 1$  integer

## exercise 4.5

# neural networks

**neural networks** are regression/classification machine learning models inspired by the functioning of the biological neuron

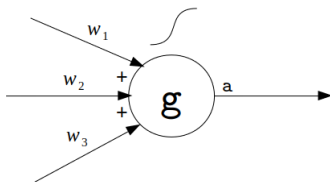


neural networks are input-output systems with a layered structure

- **input layer** propagating the input signals (features)
- **hidden layers** made up of the computational units **neurons** elaborating the signals from the previous layers
- **output layer** providing the final output signal

weighted connections link the layers

each neuron is in turn an input-output system...



- taking as input the weighted incoming signals
- elaborating their sum by means of a nonlinear **activation function**  $g$  (typically sigmoidal) to produce an output signal  $a$

training a neural network consists in tuning the weights of the connections so as to minimize a loss function representing the overall discrepancy between the outputs  $\hat{y}^i$  produced by the network for points  $x^i$  in  $TR$  and the actual outputs  $y^i$  like  $RSS = \sum_{i=1,m} (y^i - \hat{y}^i)^2$

## some neural networks considerations...

- difficult to train (training optimization problem is not convex!)
- hyperparameters are difficult to set (e.g. number of hidden layers, number of neurons on the layers, type of activation function,...)
- once trained they act as a black-box (not interpretable)
- big representative power
- can approximate any continuous function with arbitrary precision
- are the basis of **deep learning** models for e.g. image and speech recognition
- in `scikit-learn`
  - ▶ `sklearn.neural_network.MLPClassifier()` (classification model)
  - ▶ `sklearn.neural_network.MLPRegressor()` (regression model)

- linear/polynomial regression
  - ▶ 'An Introduction to Statistical Learning', Sections 3.1-3.3, 7.1
- logistic regression
  - ▶ 'An Introduction to Statistical Learning', Section 4.3
- decision trees
  - ▶ 'An Introduction to Statistical Learning', Section 8.1
- support vector machines
  - ▶ 'An Introduction to Statistical Learning', Sections 9.1-9.3
- neural networks
  - ▶ 'An Introduction to Statistical Learning', Sections 10.1-10.2
  - ▶ 'The Elements of Statistical Learning', Section 11